

Survival Analysis using SPSS

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Outline

Topics in Survival Analysis

- What is survival analysis?
- Kaplan Meier survival analysis
- Cox proportional hazards regression
- Example with SPSS Syntax
- Conclusion

What is survival analysis?

- Event history analysis
- Time series analysis
- Use when the research question is about time to event, and the event is a discrete occurrence.
- Examples: time to death, time to recurrence of disease, length of stay in hospital (time to discharge), etc
- The event is recorded for each subject – it may occur or not occur (in which case it is called right censored)

1. Kaplan Meier Survival Analysis

Introduction:

This is a method of estimating time-to-event in the presence of censored cases.

The procedure describes the distribution of time-to-event variables.

We typically want to compare the distribution by levels of a factor (eg different drug treatments).

1. Kaplan Meier Survival Analysis

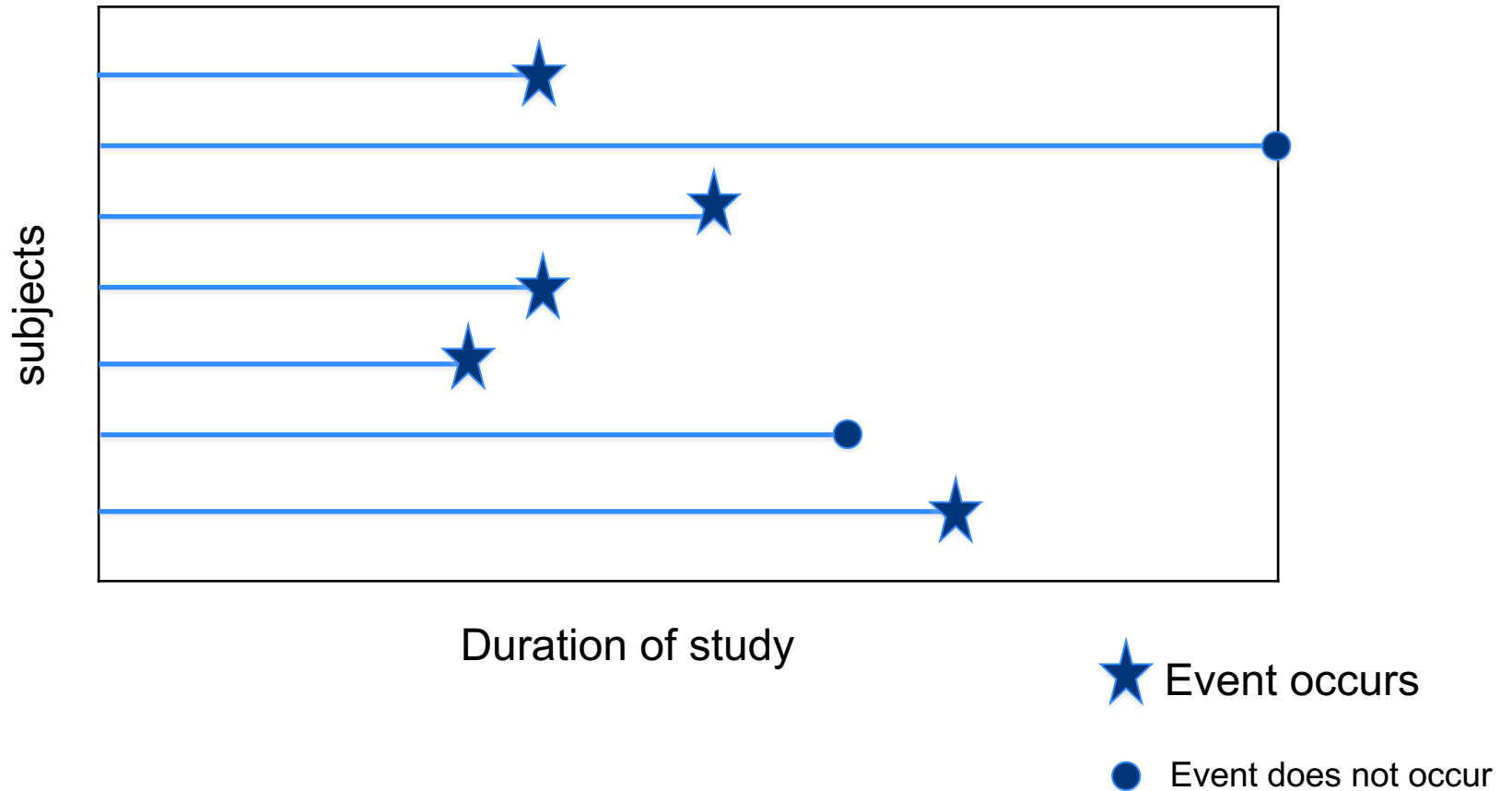
Assumptions of K-M survival analysis

- Probabilities for the event should only depend on time (since the start point). No covariate effects are considered.
- Cases that enter the study at different times behave similarly.
- Probability of the event occurring is not related to censoring.

What could go wrong with these assumptions?

1. Kaplan Meier Survival Analysis

Censoring



1. Kaplan Meier Survival Analysis

SPSS Case Study:

A pharmaceutical company is developing an anti-inflammatory medication for treating chronic arthritic pain. Of particular interest is the time it takes for the drug to take effect and how it compares to an existing medication.

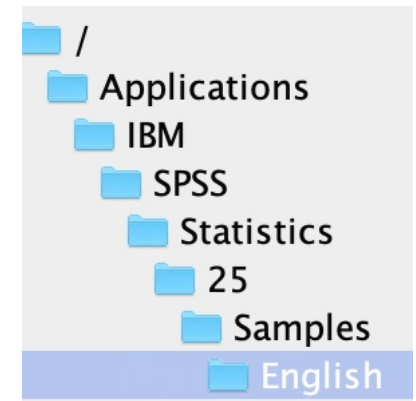
Shorter times to effect are considered better.

The results of a clinical trial are collected in *pain_medication.sav*.

Use Kaplan-Meier Survival Analysis to examine the distribution of "time to effect" and compare the effectiveness of the two treatments.

1. Kaplan Meier Survival Analysis

1. Open SPSS
2. Open data > choose “**pain_medication**”
3. Go to help/topics, then search “pain relief”
4. Choose “Using Kaplan-Meier Survival Analysis to Test Competing Pain Relief Treatments”



+ Using Kaplan-Meier Survival Analysis to Test Competing **Pain Relief** Treatments

A pharmaceutical company is developing an anti-inflammatory medication for treating chronic arthritic **pain**. Of particular interest is the time it takes for the drug to take effect and how it compares to an existing medication

1. Kaplan Meier Survival Analysis

Step 0: Not in the tutorial

Look at the data!

What should we look at?

1. Kaplan Meier Survival Analysis

Step 0: Not in the tutorial

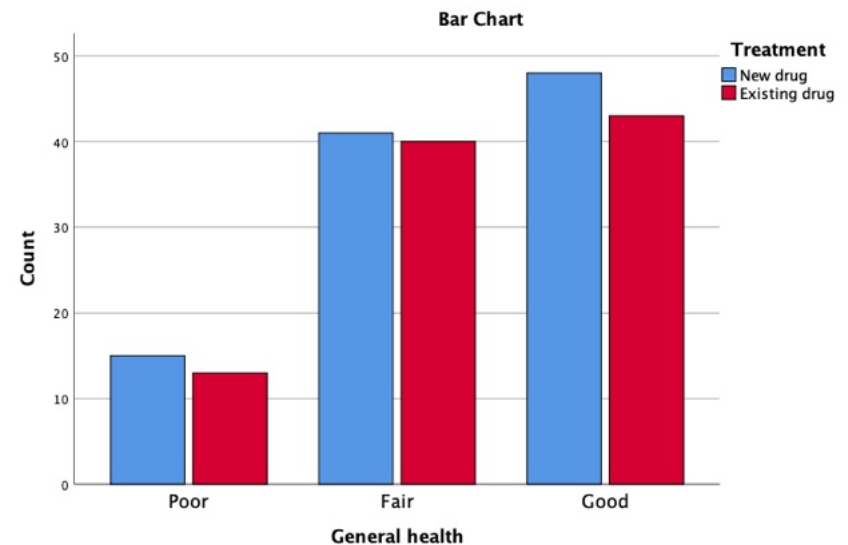
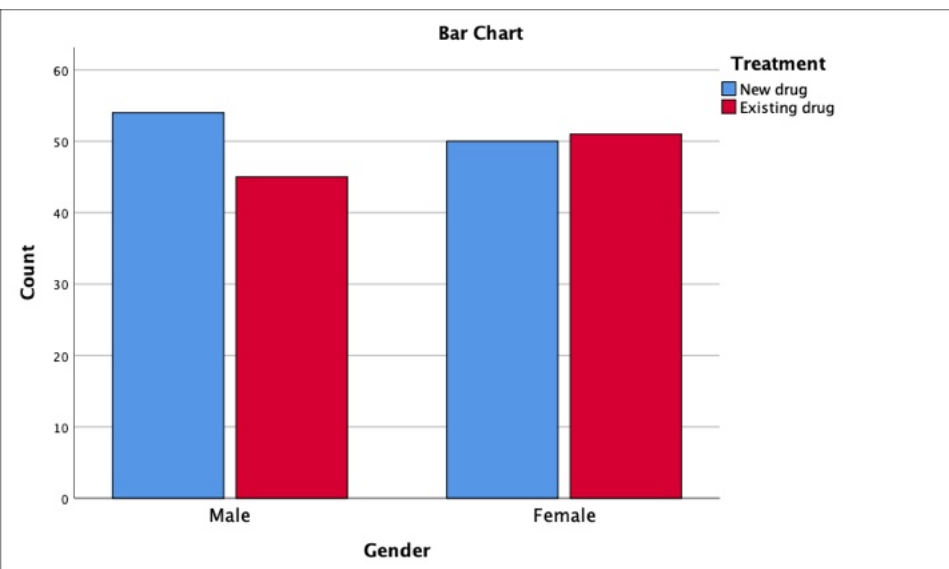
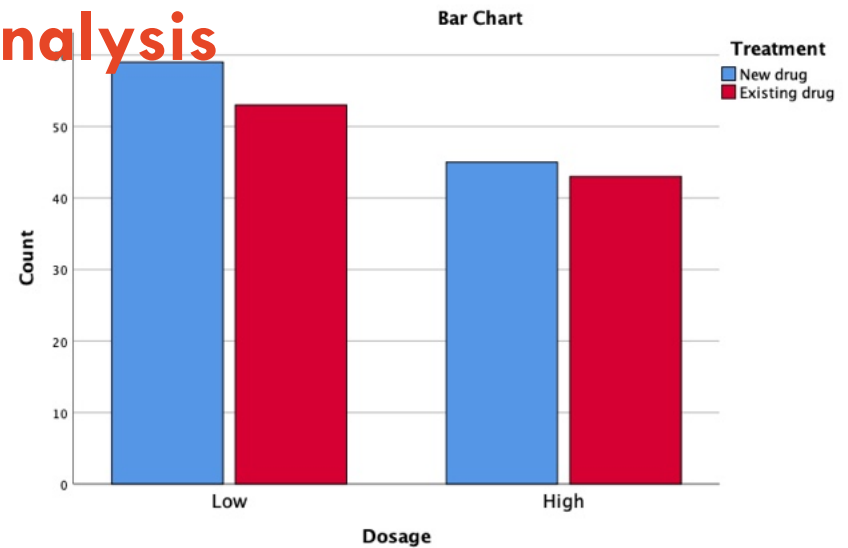
event
↓
Time to event
↓

	 age	 gender	 health	 treatment	 dosage	 status	 time
1	54	Male	Fair	New drug	High	Taken effect	1.20
2	54	Male	Poor	New drug	High	Taken effect	4.00
3	64	Male	Good	New drug	High	Taken effect	7.40
4	63	Male	Good	Existing drug	High	Taken effect	7.30
5	43	Male	Good	Existing drug	High	Censored	7.40
6	67	Female	Poor	New drug	High	Taken effect	.60
7	45	Female	Fair	New drug	Low	Taken effect	1.90
8	49	Female	Good	Existing drug	Low	Taken effect	2.70
9	59	Female	Good	New drug	Low	Taken effect	1.10
10	56	Male	Poor	Existing drug	Low	Taken effect	1.30

1. Kaplan Meier Survival Analysis

Exploratory Data Analysis

Use “crosstabs” to produce tables and plots of categorical variable frequencies.

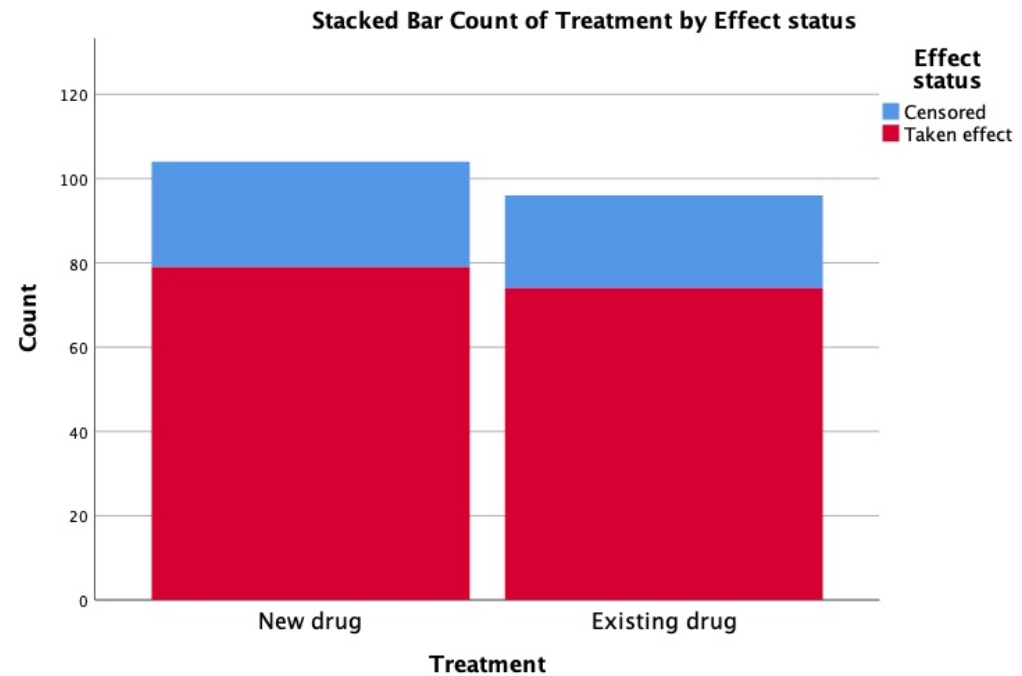


1. Kaplan Meier Survival Analysis

Exploratory Data Analysis

How many cases?

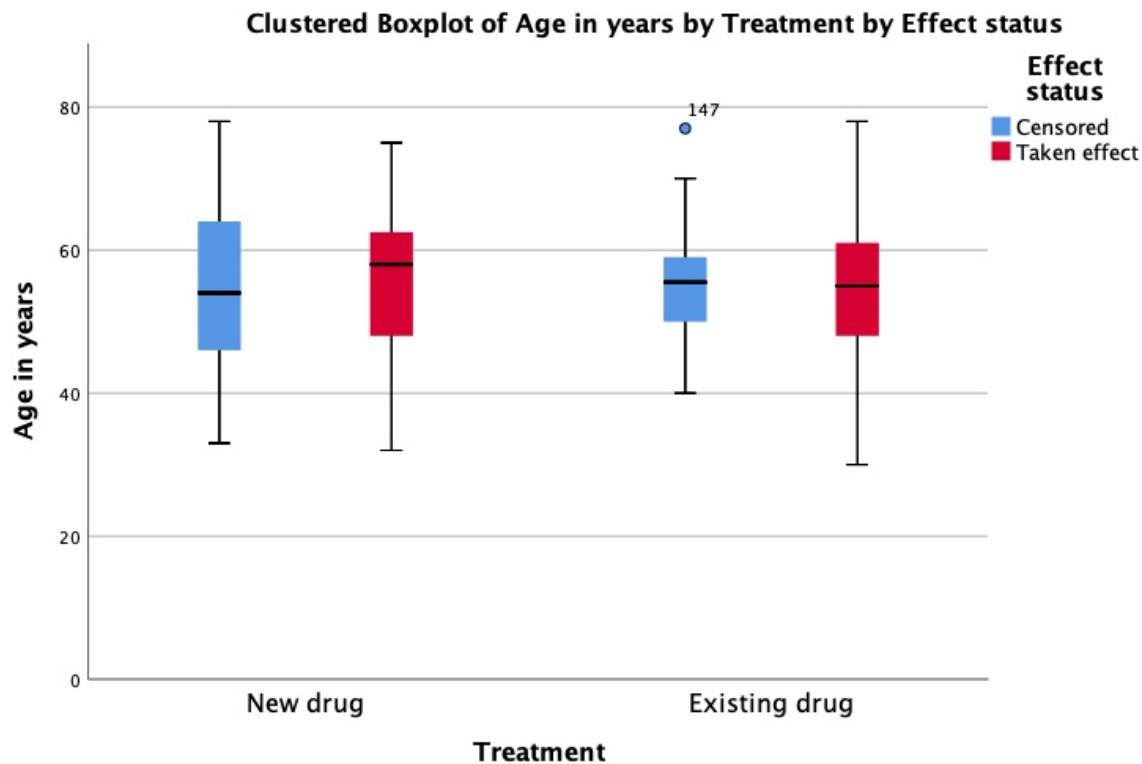
How many are censored?



1. Kaplan Meier Survival Analysis

Exploratory Data Analysis

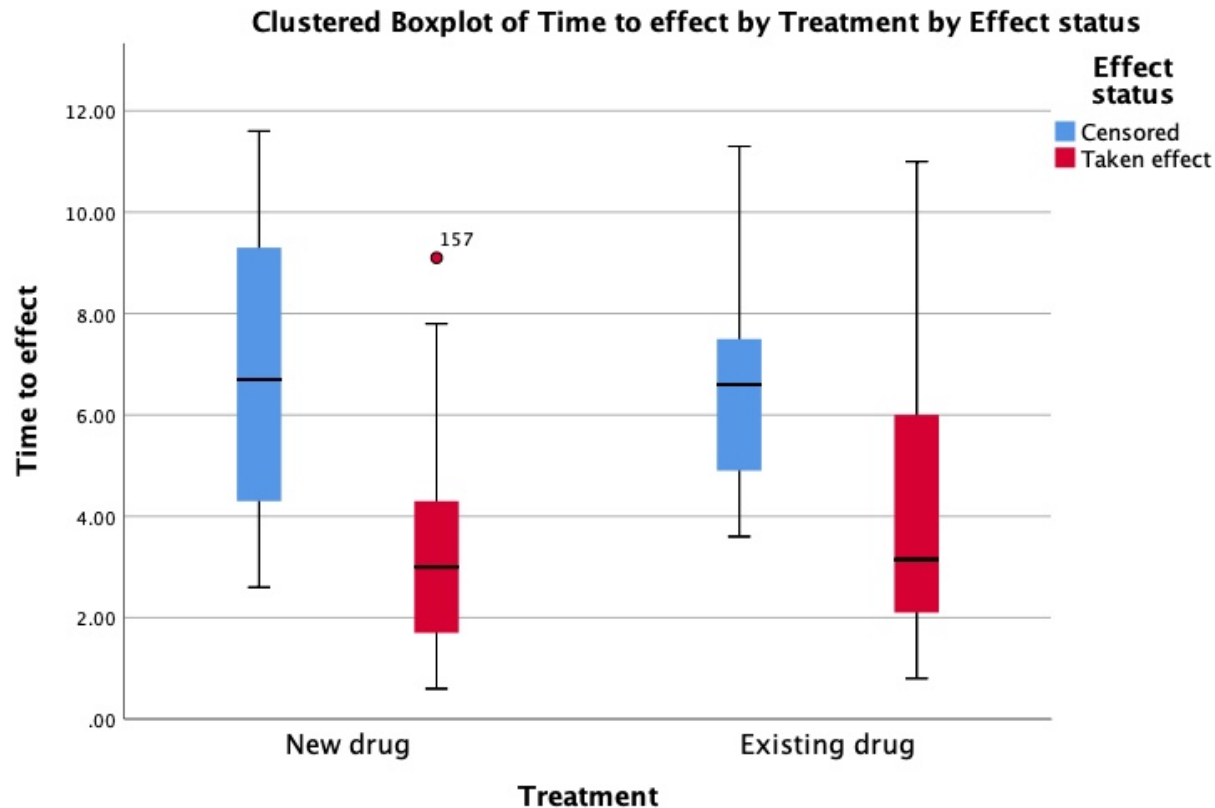
Use “chart builder” to produce plots of continuous variables by group.



1. Kaplan Meier Survival Analysis

Exploratory Data Analysis

Use “chart builder” to produce plots of continuous variables by group.



1. Kaplan Meier Survival Analysis

Exploratory Data Analysis

In this case the EDA has not revealed any concerns regarding the balance of the groups in terms of covariates.

The frequency of censoring is not too high.

It also shows that there are no concerns with outliers, missing values, or anything unexpected.

1. Kaplan Meier Survival Analysis

Now follow the steps in the tutorial.

Step 1: Prepare the analysis

1. To run a Kaplan-Meier Survival Analysis, from the menus choose:
Analyze > Survival > Kaplan-Meier...

Figure 1. Kaplan-Meier Survival Analysis main dialog box



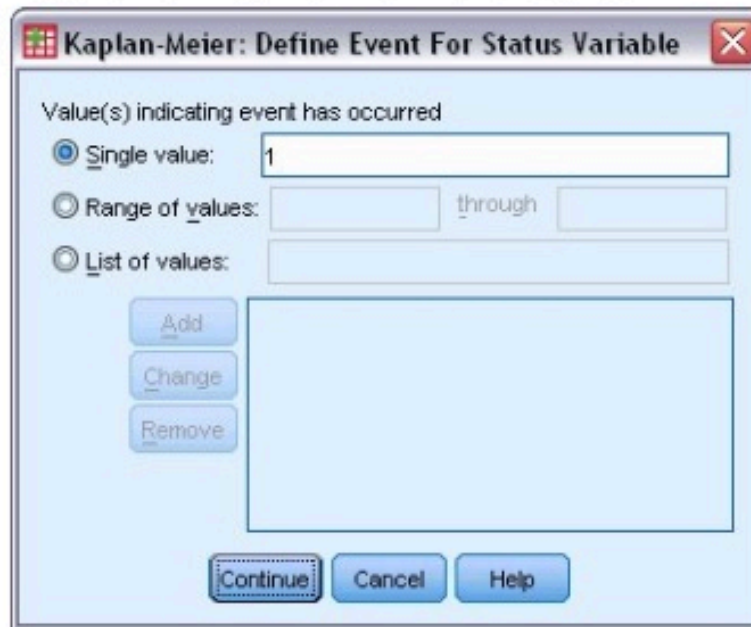
2. Select *Time to effect* as the time variable.
3. Select *Effect status* as the status variable.

1. Kaplan Meier Survival Analysis

Step 1: Prepare the analysis

4. Click **Define Event**.

Figure 2. Define Event dialog box



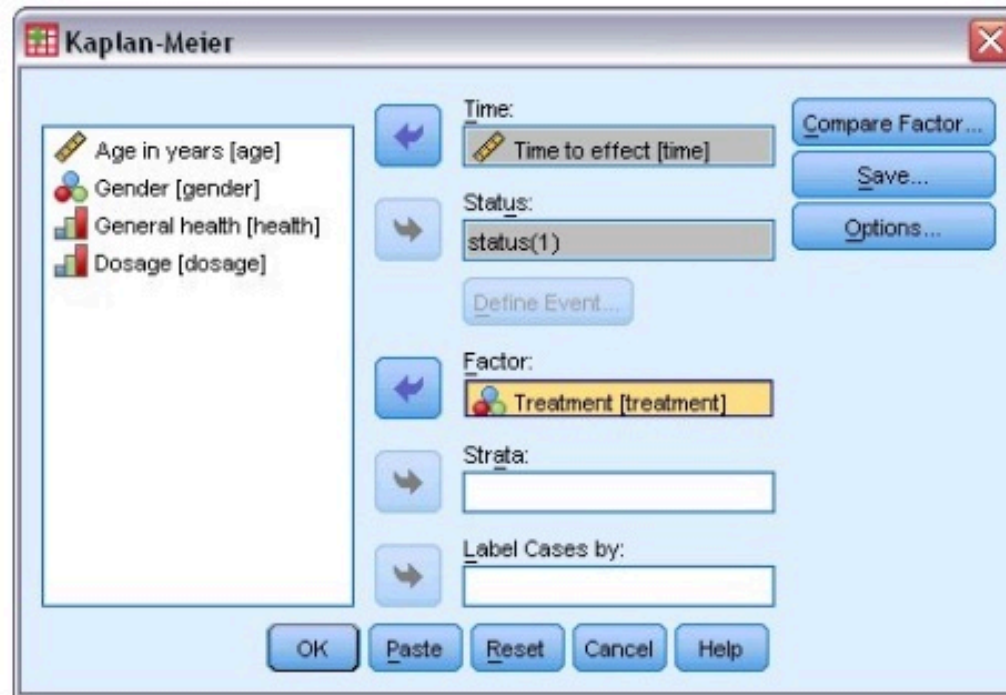
5. In the Define Event dialog, type 1 as the single value indicating the medication took effect.

6. Click **Continue**.

1. Kaplan Meier Survival Analysis

Step 1: Prepare the analysis

Figure 3. Kaplan-Meier Survival Analysis main dialog box



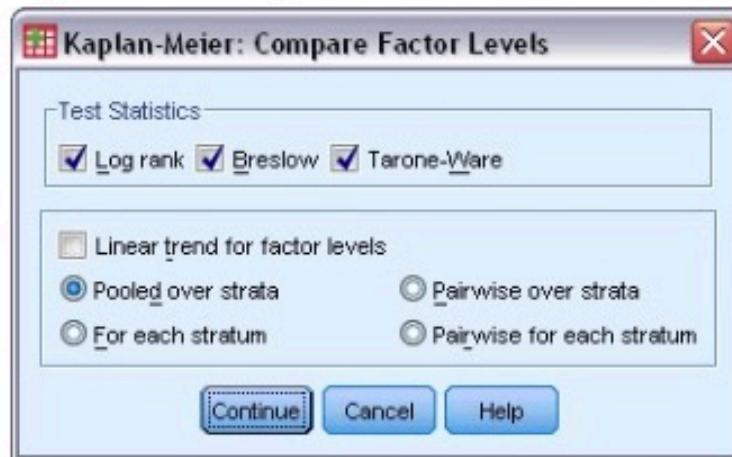
7. In the Kaplan-Meier Survival Analysis dialog, select *Treatment* as a factor.

1. Kaplan Meier Survival Analysis

Step 1: Prepare the analysis

8. Click **Compare Factor**.

Figure 4. Compare Factor Levels dialog box



9. In the Compare Factor Levels dialog, select **Log rank**, **Breslow**, and **Tarone-Ware**.

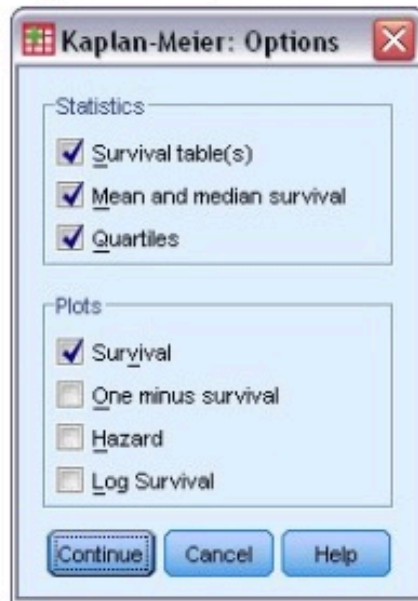
10. Click **Continue**.

1. Kaplan Meier Survival Analysis

Step 1: Prepare the analysis

11. Click **Options** in the Kaplan-Meier Survival Analysis dialog box.

Figure 5. Options dialog box



12. In the Options dialog, select **Quartiles** in the Statistics group and **Survival** in the Plots group.
13. Click **Continue**.
14. Click **OK** in the Kaplan-Meier Survival Analysis dialog box.

1. Kaplan Meier Survival Analysis

Step 2: View and consider the output

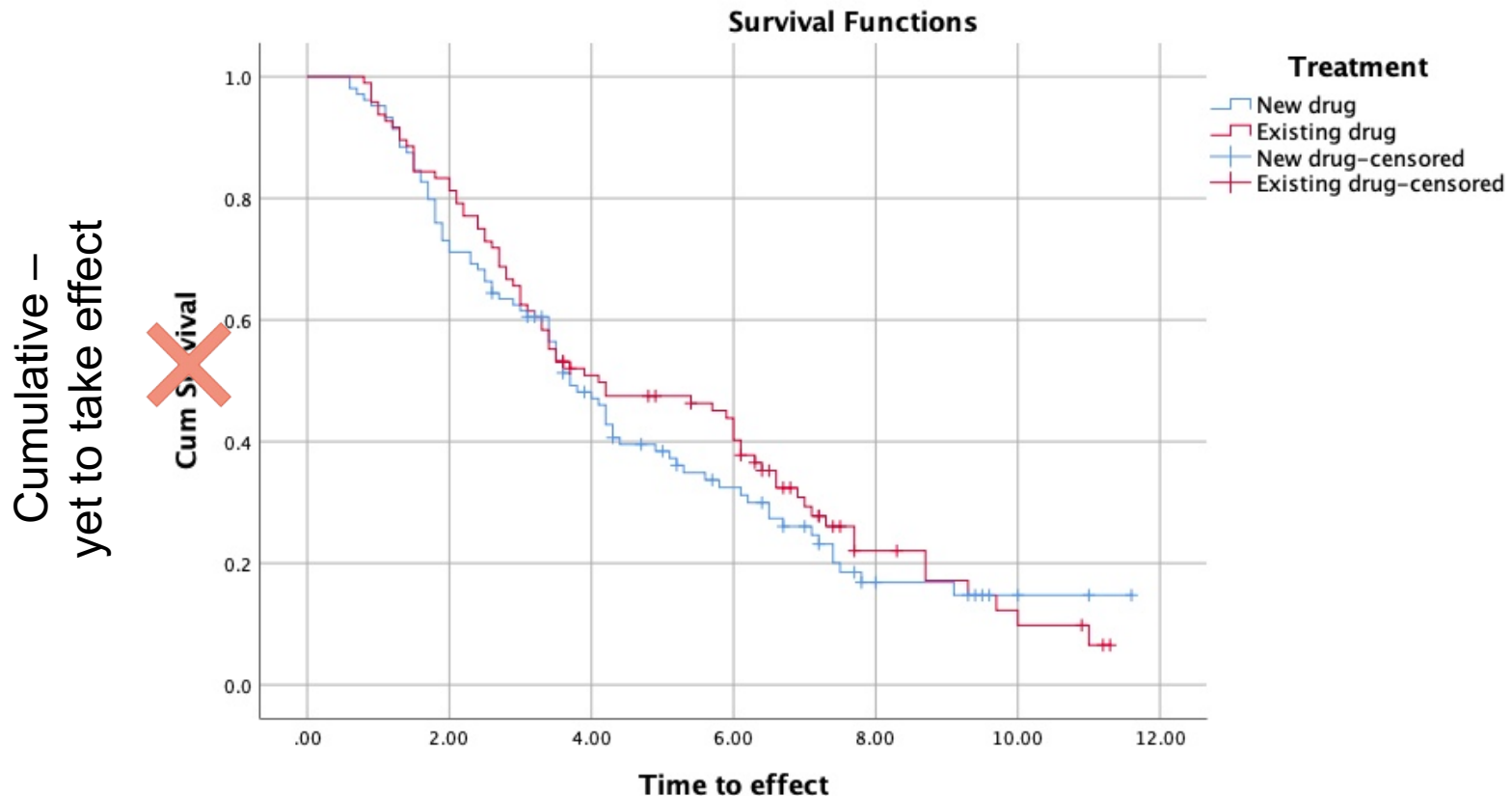
Survival Table

Treatment		Time	Status	Cumulative Proportion Surviving at the Time		N of Cumulative Events	N of Remaining Cases
				Estimate	Std. Error		
New drug	1	.600	Taken effect	.	.	1	103
	2	.600	Taken effect	.981	.013	2	102
	3	.700	Taken effect	.971	.016	3	101
	4	.800	Taken effect	.962	.019	4	100
	5	.900	Taken effect	.952	.021	5	99
	6	1.100	Taken effect	.	.	6	98
	7	1.100	Taken effect	.933	.025	7	97
	8	1.200	Taken effect	.	.	8	96
	9	1.200	Taken effect	.913	.028	9	95
	10	1.300	Taken effect	.	.	10	94

1. Kaplan Meier Survival Analysis

Step 2: View and consider the output

The survival curves give a visual representation of the life tables.



1. Kaplan Meier Survival Analysis

Step 2: View and consider the output

Figure 1. Means and medians for survival time

treatment	Mean ^a				Median			
	Estimate	Std. Error	95% Confidence Interval		Estimate	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound			Lower Bound	Upper Bound
New drug	4.867	.360	4.162	5.572	3.700	.292	3.128	4.272
Existing drug	5.185	.350	4.499	5.871	4.100	1.131	1.884	6.316
Overall	4.905	.478	3.968	5.841	3.700	.320	3.072	4.328

a. Estimation is limited to the largest survival time if it is censored.

The means and medians for survival time table offers a quick numerical comparison of the "typical" times to effect for each of the medications. Since there is a lot of overlap in the confidence intervals, it is unlikely that there is much difference in the "average" survival time.

1. Kaplan Meier Survival Analysis

Step 2: View and consider the output

Figure 2. Pairwise comparisons

treatment	25.0%		50.0%		75.0%	
	Estimate	Std. Error	Estimate	Std. Error	Estimate	Std. Error
New drug	7.100	.509	3.700	.292	1.900	.226
Existing drug	7.700	.648	4.100	1.131	2.400	.247
Overall	7.100	.224	3.700	.292	1.900	.154

The percentiles table gives estimates of the first quartile, median, and third quartile of the survival distribution. The interpretation of percentiles for survival curves is that the 75th percentile is the latest time that at least 75 percent of the patients have yet to feel relief.

1. Kaplan Meier Survival Analysis

Step 2: View and consider the output

Figure 3. Overall comparisons

	Chi-Square	df	Sig.
Log Rank (Mantel-Cox)	.379	1	.538
Breslow (Generalized Wilcoxon)	.748	1	.387
Tarone-Ware	.705	1	.401

Test of equality of survival distributions for the different levels of treatment.

This table provides overall tests of the equality of survival times across groups. Since the significance values of the tests are all greater than 0.10, you cannot determine a difference between the survival curves.

1. Kaplan Meier Survival Analysis

Step 2: View and consider the output

Summary

With the Kaplan-Meier Survival Analysis procedure, you have examined the distribution of time to effect for two different medications.

The comparison tests show that there is not a statistically significant difference between them.

1. Kaplan Meier Survival Analysis

Step 3: What to report?

- Type of analysis: Kaplan-Meier
- Descriptive Statistics – EDA
- output

1. Kaplan Meier Survival Analysis

Step 3: What to report?

- Tidy up tables – alignment, decimal places, etc.
- Double-click table > select cells > right-click > Cell properties

Means and Medians for Survival Time

Treatment	Mean ^a				Median			
	Estimate	Std. Error	95% Confidence Interval		Estimate	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound			Lower Bound	Upper Bound
New drug	4.9	.4	4.2	5.6	3.7	.3	3.1	4.3
Existing drug	5.2	.4	4.5	5.9	4.1	1.1	1.9	6.3
Overall	5.0	.3	4.5	5.5	3.9	.3	3.4	4.4

a. Estimation is limited to the largest survival time if it is censored.

1. Survival

To survive you have to keep going!



2. Cox Regression Analysis

Introduction:

- Cox regression is used to model the time-to-event based on one or more covariates.
- We can (try) to predict the event occurring based on the values of one or more covariates.
- We estimate Hazard Ratios for the factors and covariates in the model.
- This is like logistic regression where we estimate odds ratios.

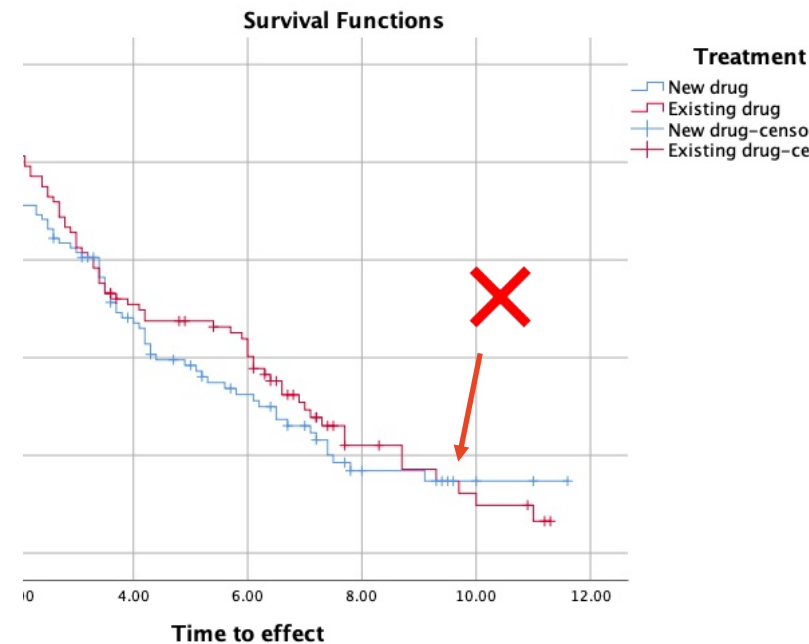
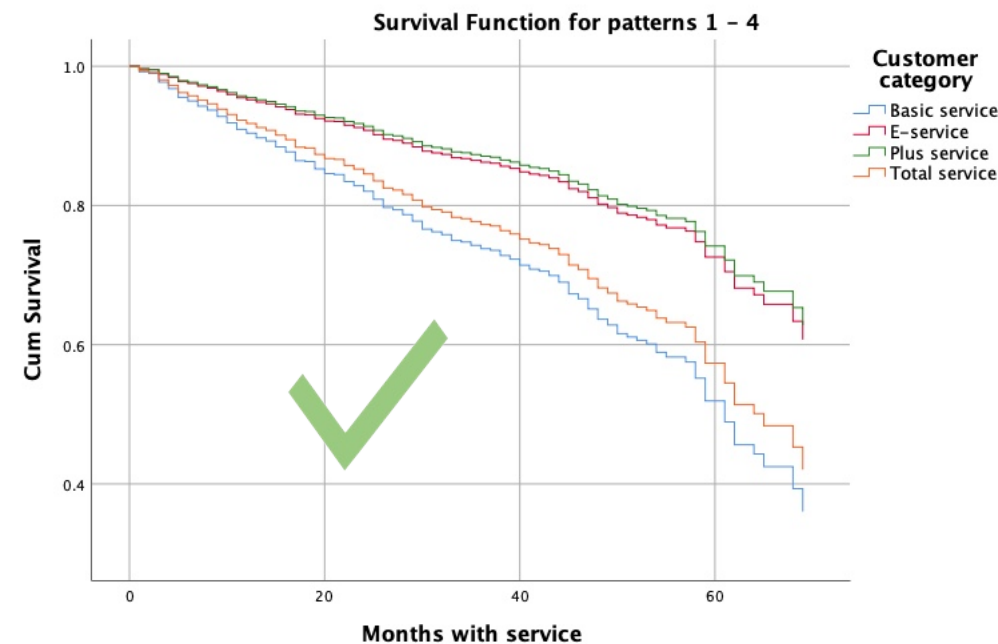
2. Cox Regression Analysis

Assumptions of Cox Regression:

Proportional hazards assumption (very important!)

The ratio of the hazards for any two subjects is constant over time.

Check by looking at the survival curves



2. Cox Regression Analysis

Other assumptions...

Linear covariate relationships

Each covariate makes a linear contribution to the model (correct functional form of covariates)

Independence

The probability of the event occurring to any subject is independent of (or not correlated with) any other subject.

2. Cox Regression Analysis

SPSS Case Study:

Using Cox Regression to model customer “time to churn”

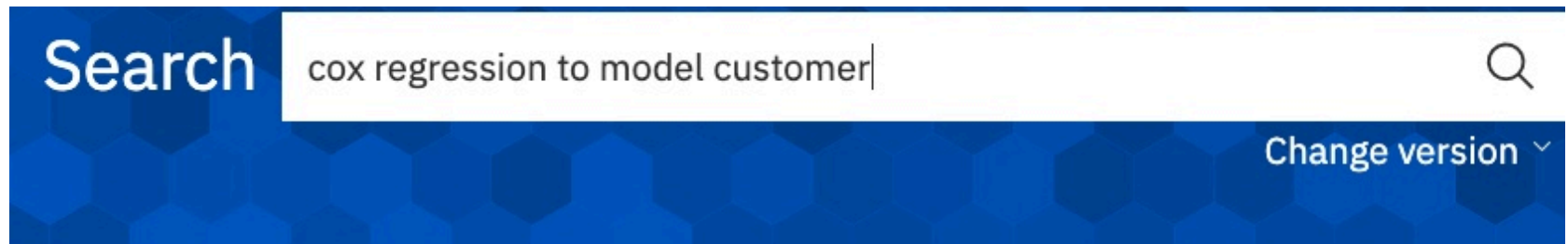


As part of its efforts to reduce customer churn, a telecommunications company is interested in modeling the "time to churn" in order to determine the factors that are associated with customers who are quick to switch to another service. To this end, a random sample of customers is selected and their time spent as customers, whether they are still active customers, and various demographic fields are pulled from the database.

We are especially interested in the relationship between “customer category” and churn.

2. Cox Regression Analysis

1. Open SPSS
2. Open another file > choose “telco.sav”
3. Go to help/topics, then search “cox regression to model customer”
4. Choose “Using Cox Regression to model customer time to churn”



Documentation

Videos

IBM Developer

Technotes

Redbooks

198 results

+ Using Cox Regression to Model Customer Time to Churn

As part of its efforts to reduce **customer** churn, a telecommunications company is interested in **modeling** the "time to churn" in order to determine the factors that are associated with **customers** who are quick to switch to another service. To this end

2. Cox Regression Analysis

Step 0: not in the SPSS tutorial

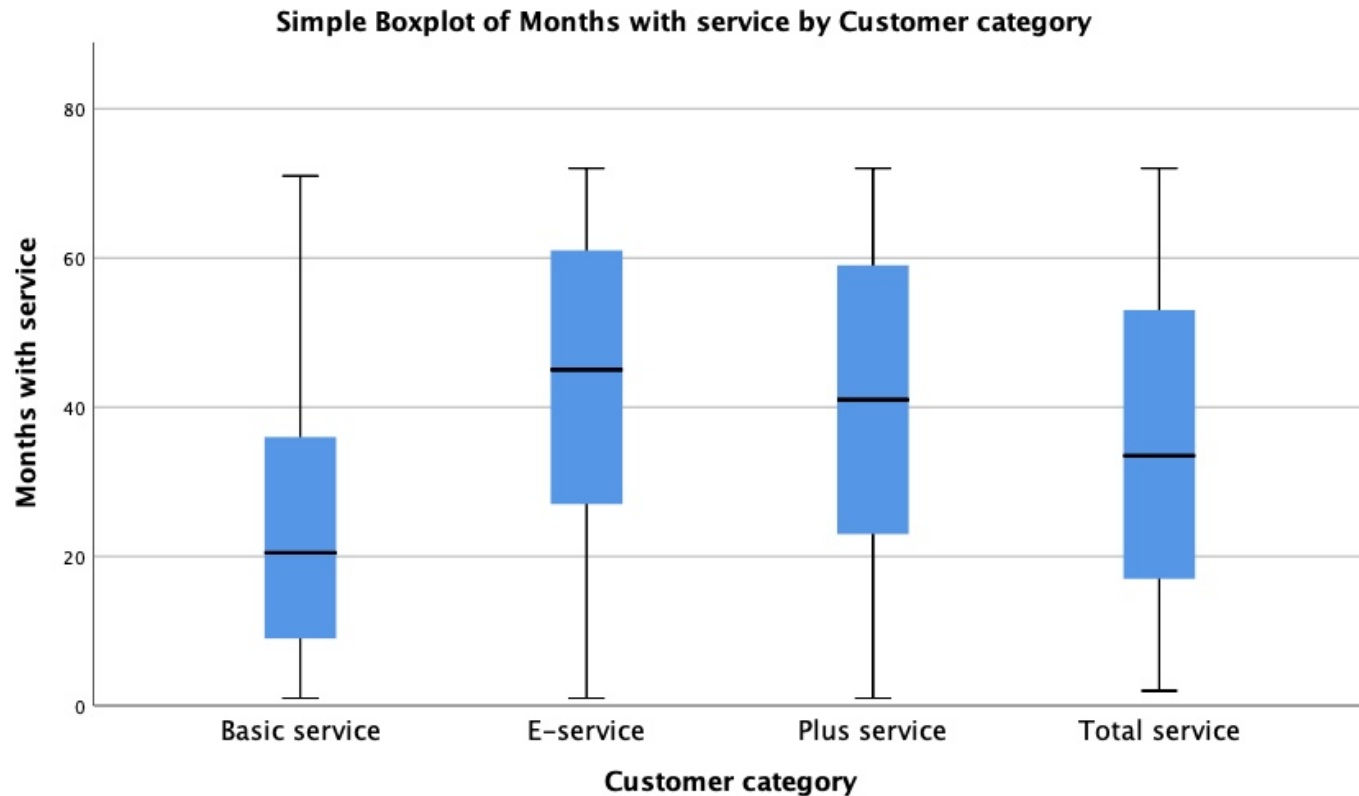
Look at the data!

What should we look at?

2. Cox Regression Analysis

Step 0: not in the SPSS tutorial

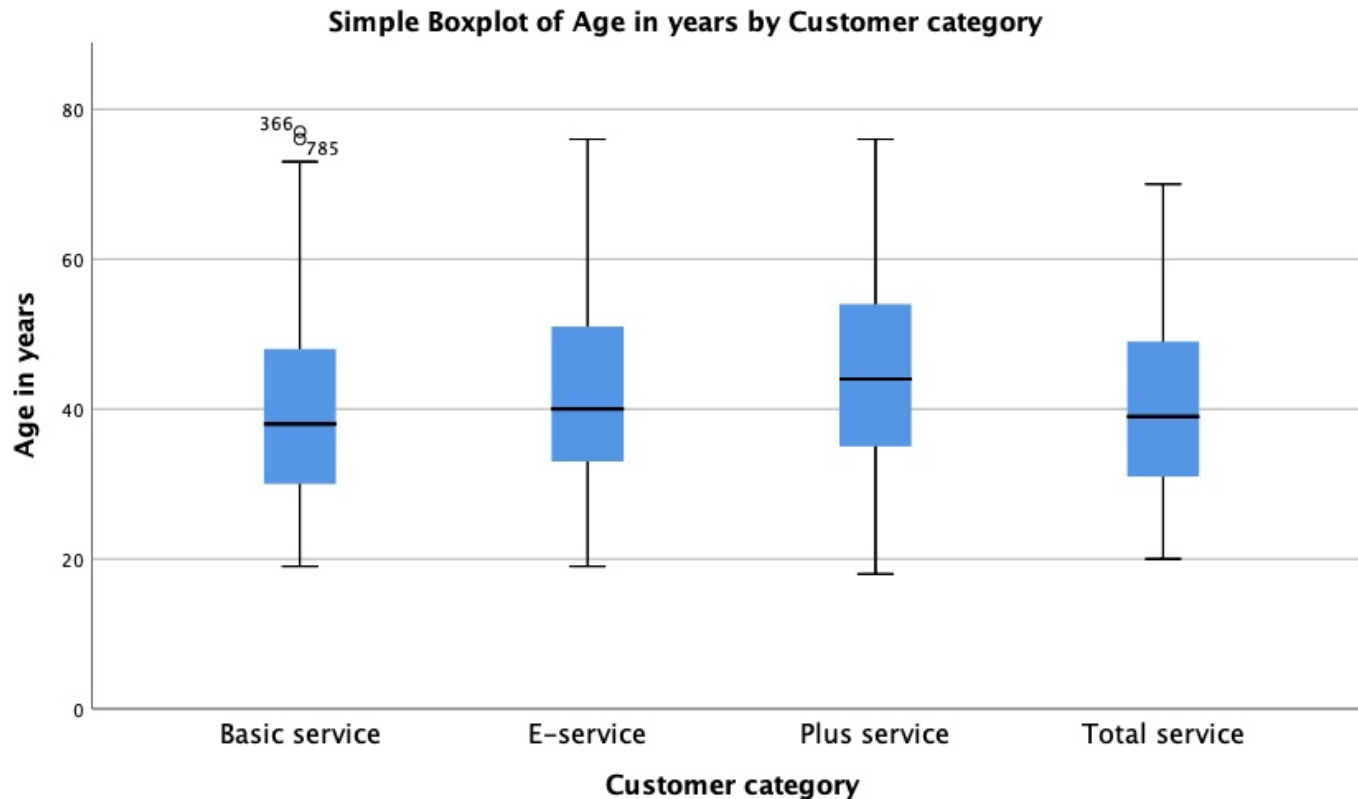
Exploratory Data Analysis



2. Cox Regression Analysis

Step 0: not in the SPSS tutorial

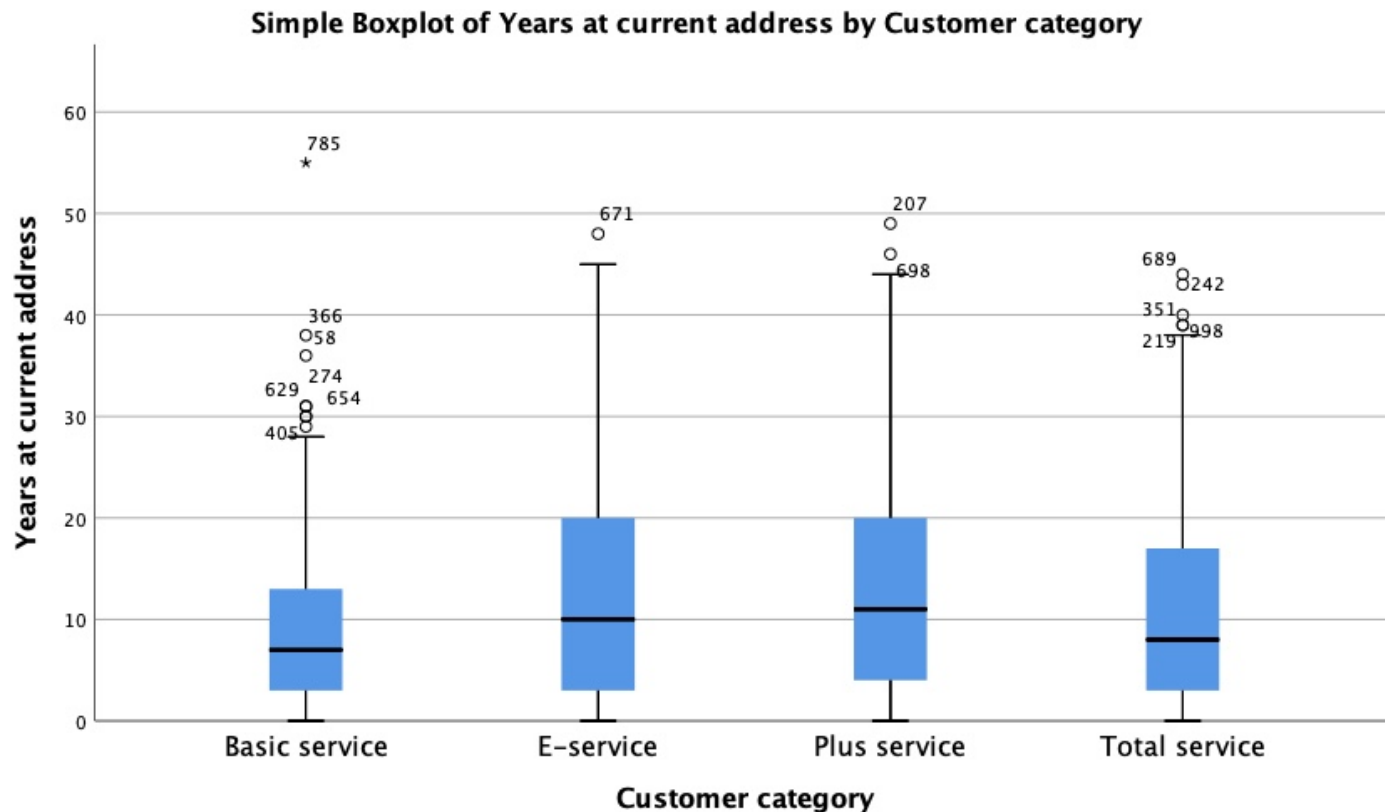
Exploratory Data Analysis



2. Cox Regression Analysis

Step 0: not in the SPSS tutorial

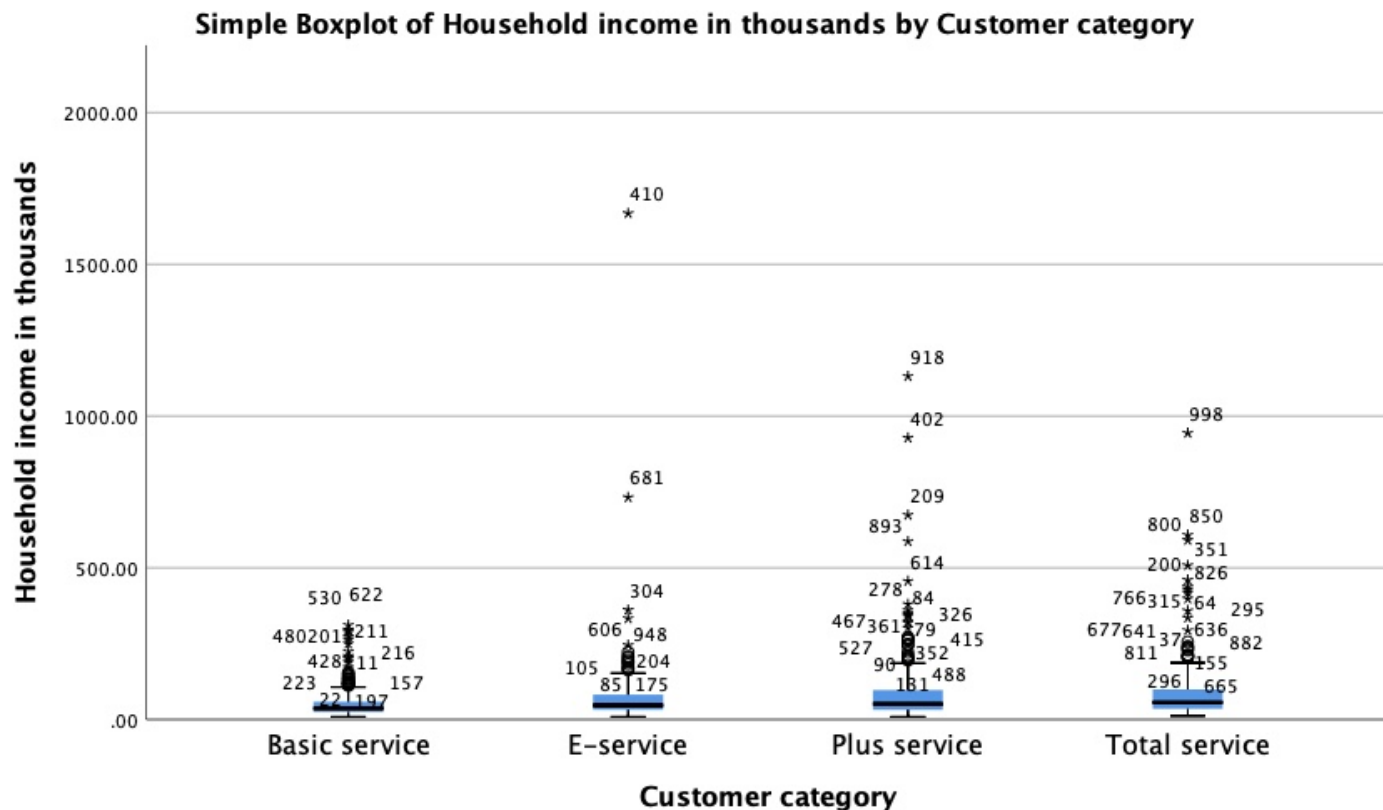
Exploratory Data Analysis



2. Cox Regression Analysis

Step 0: not in the SPSS tutorial

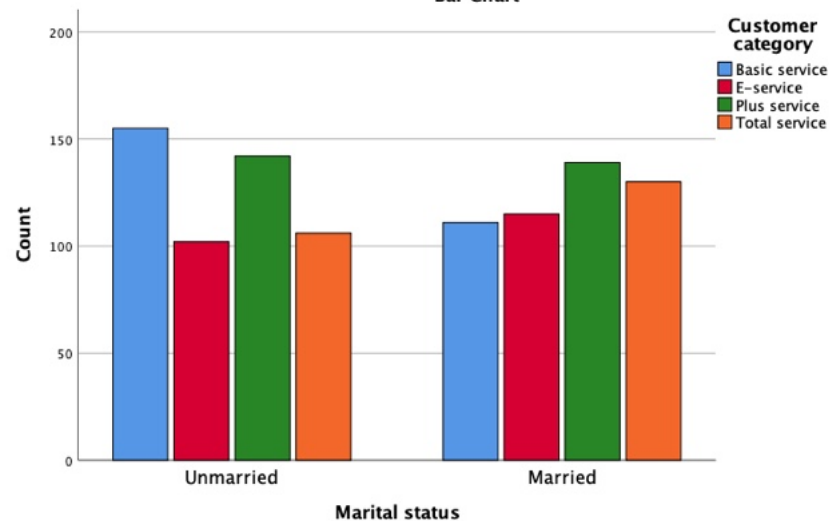
Exploratory Data Analysis



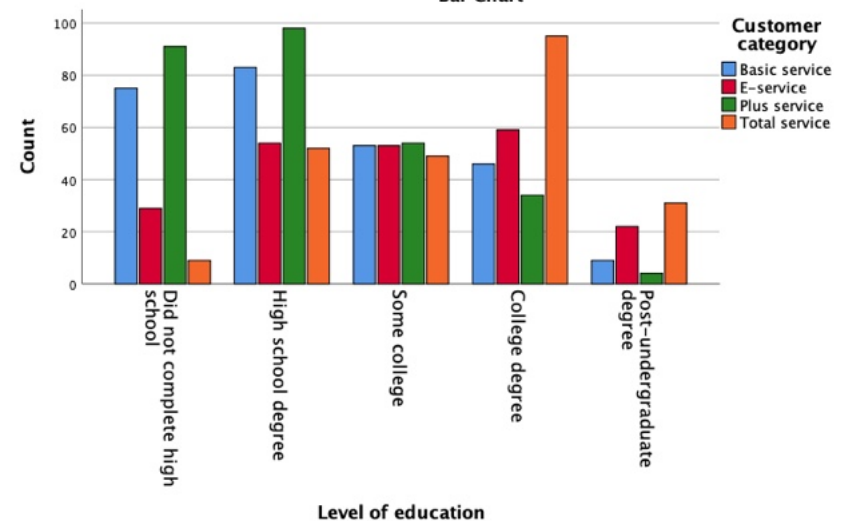
2. Cox Regression Analysis

Step 0: not in the SPSS tutorial
Exploratory Data Analysis

Bar Chart

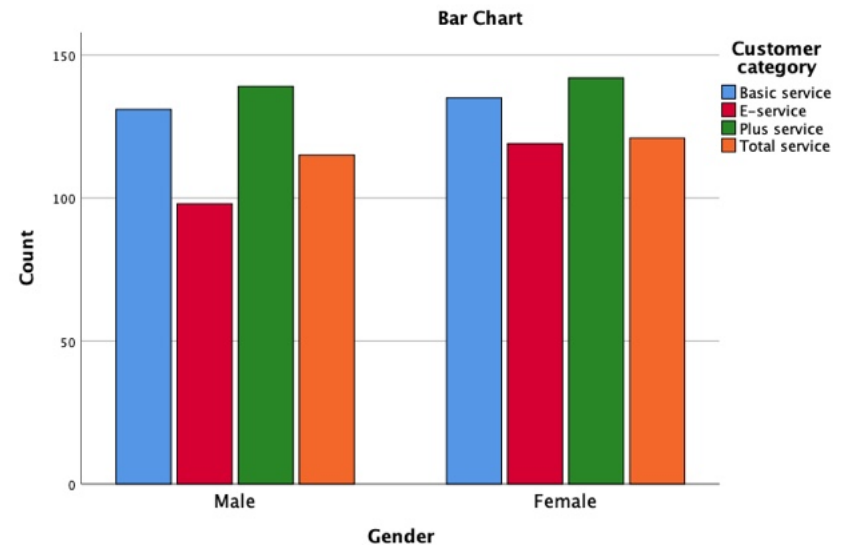
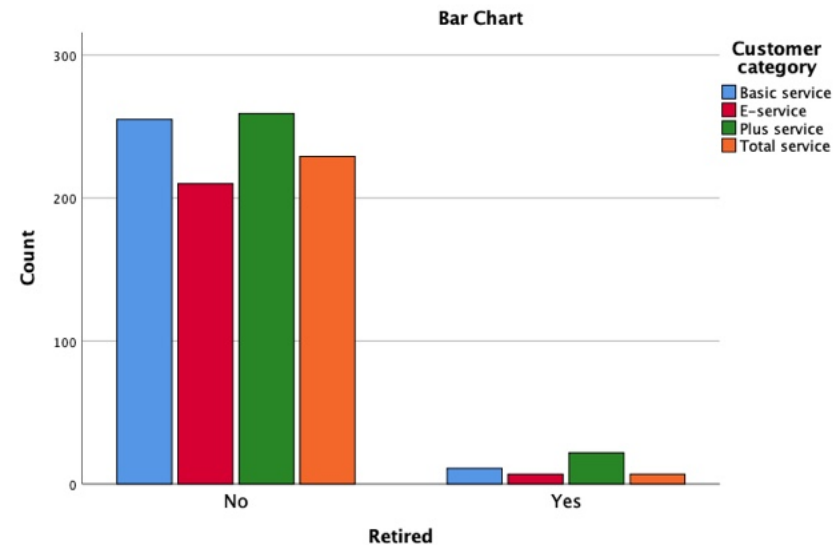


Bar Chart



2. Cox Regression Analysis

Step 0: not in the SPSS tutorial
Exploratory Data Analysis



2. Cox Regression Analysis

Step 0: not in the SPSS tutorial

Exploratory Data Analysis –what else?

Have a look at the frequency of events

Analyze>Frequencies

Churn within last month

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	726	72.6	72.6	72.6
	Yes	274	27.4	27.4	100.0
	Total	1000	100.0	100.0	

Censored cases = 72.6%

2. Cox Regression Analysis

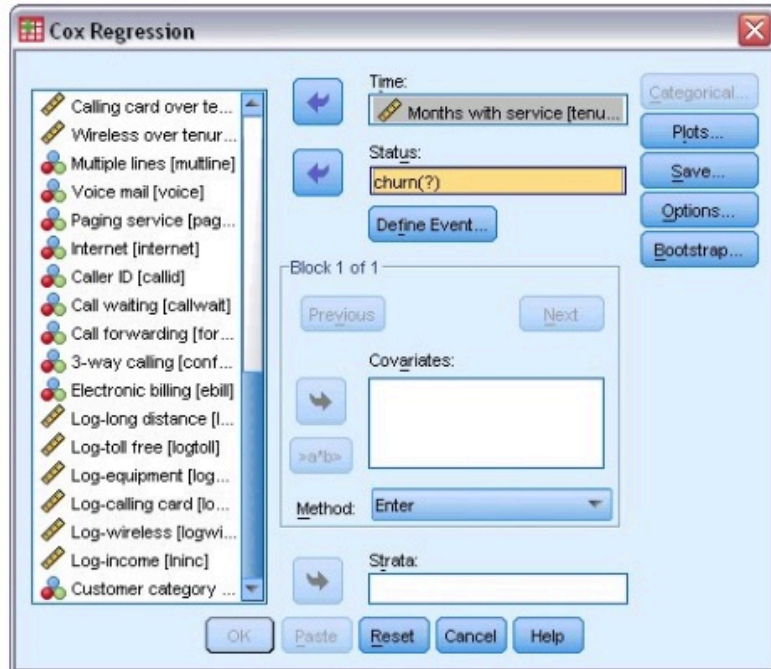
Now follow the steps in the tutorial.

Step 1: Prepare the analysis

1. To run a Cox Regression analysis, from the menus choose:

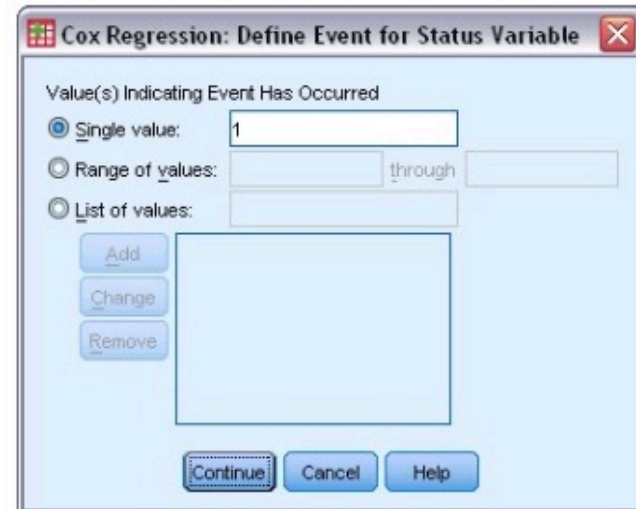
Analyze > Survival > Cox Regression...

Figure 1. Cox Regression main dialog box



4. Click **Define Event**.

Figure 2. Define Event dialog box



5. In the Define Event dialog, type 1 as the single value indicating the customer churned.
6. Click **Continue**.

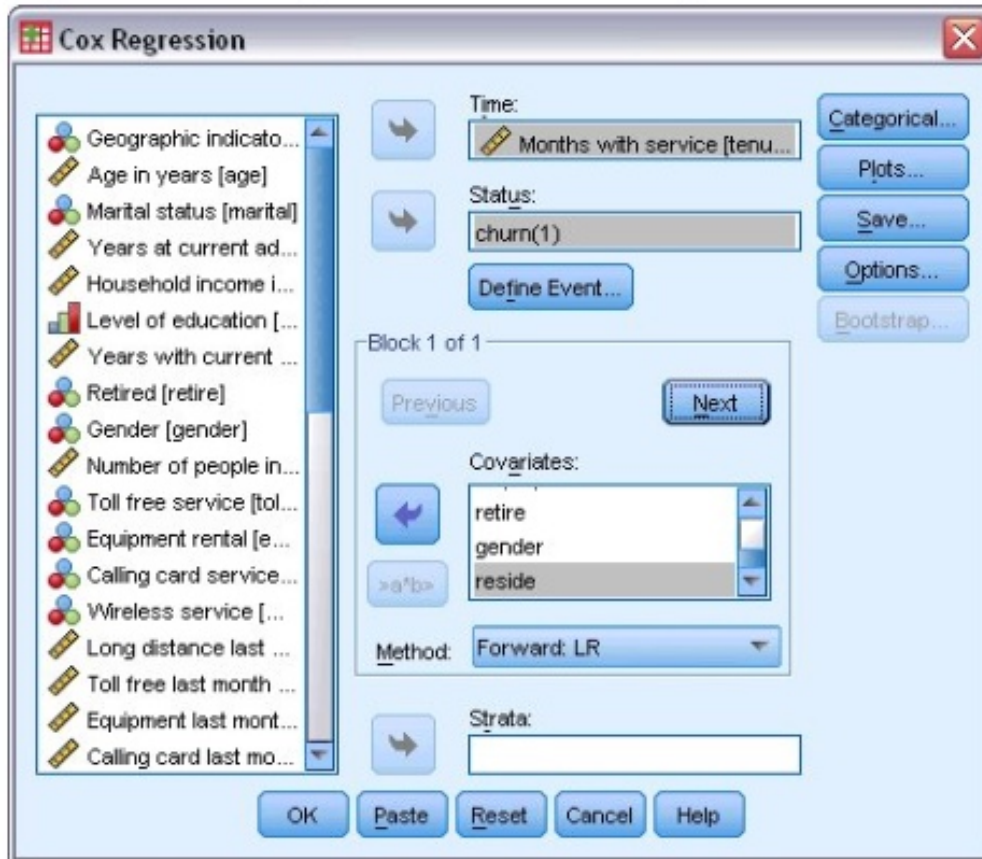
2. Select *Months with service [tenure]* as the time variable.

3. Select *Churn within last month [churn]* as the status variable.

2. Cox Regression Analysis

Step 1: Prepare the analysis

Figure 3. Cox Regression main dialog box



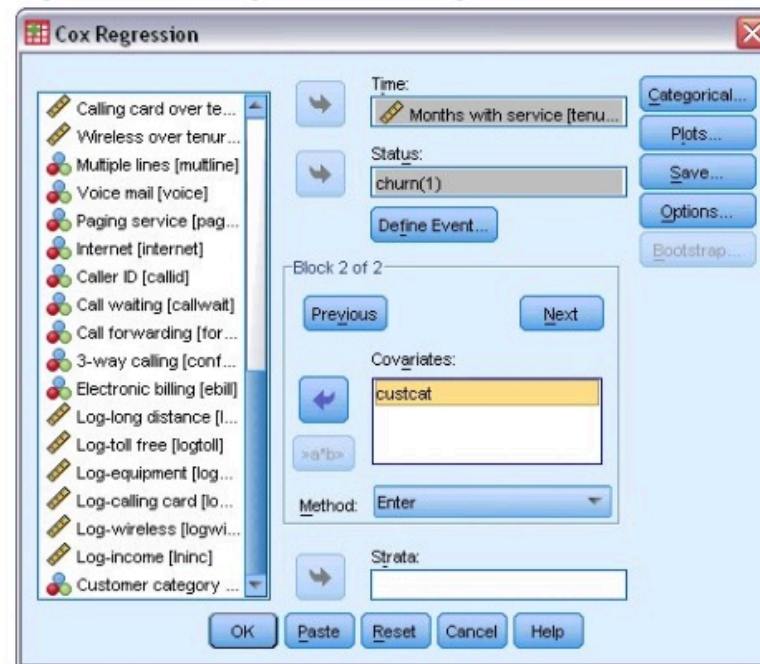
7. In the Cox Regression dialog, and with variables listed in file order, select *Age in years [age]* through *Years at current address [address]* and *Level of education [ed]* through *Number of people in household [reside]* as covariates.
8. Select **Forward: LR** in the Method drop-down list.

2. Cox Regression Analysis

Step 1: Prepare the analysis

9. Click **Next** in the Block group.

Figure 4. Cox Regression dialog box



10. Select *Customer category* [*custcat*] as a covariate.

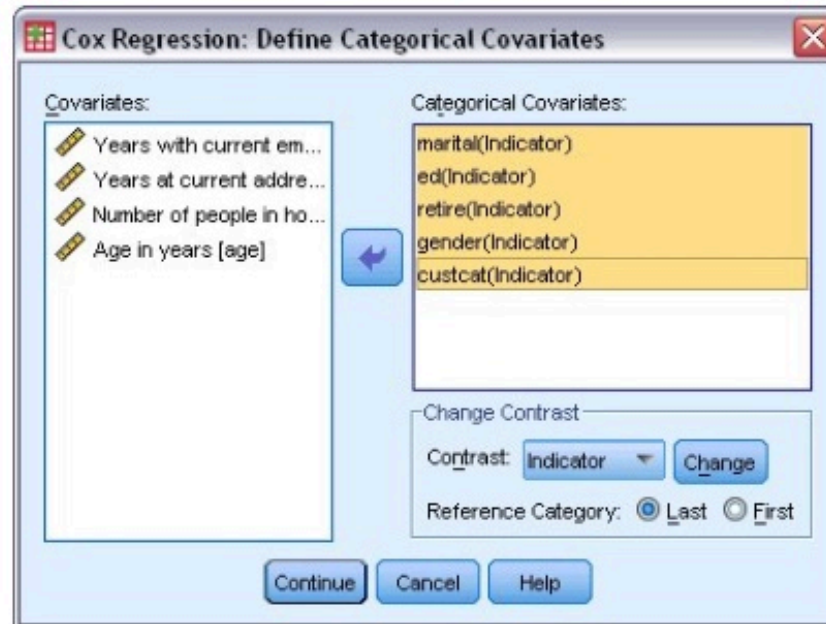
Leave the default value of Enter in the Method drop-down list. By using the entry method in a second block, this guarantees that *Customer category* will be included in the model.

2. Cox Regression Analysis

Step 1: Prepare the analysis

11. Click **Categorical**.

Figure 5. Categorical Covariate dialog box



12. In the Categorical Covariates dialog, select *marital*, *ed*, *retire*, *gender*, and *custcat* as categorical covariates.

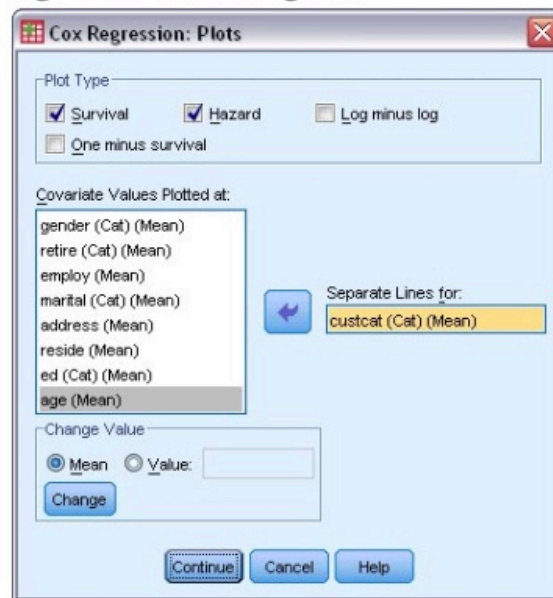
13. Click **Continue**.

2. Cox Regression Analysis

Step 1: Prepare the analysis

14. Click **Plots** in the Cox Regression dialog box.

Figure 6. Plots dialog box



15. In the Plots dialog, select **Survival** and **Hazard** in the Plot Type group.

16. Choose to have separate lines produced for each level of *custcat*.

17. Click **Continue**.

18. Click **OK** in the Cox Regression dialog box.

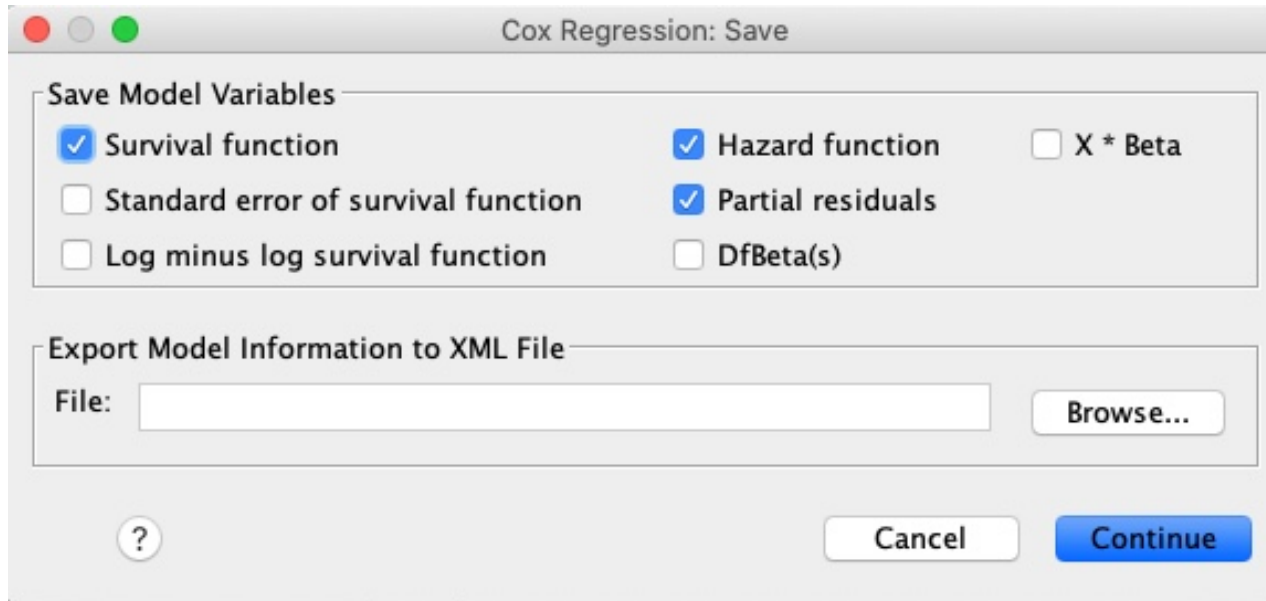
Wait!

2. Cox Regression Analysis

Step 1: Prepare the analysis

Additional step, not in the SPSS tutorial.

Save:



The image shows a screenshot of the 'Cox Regression: Save' dialog box in SPSS. The dialog box has a title bar with standard macOS window controls (red, yellow, green buttons). The main content area is divided into two sections. The first section, 'Save Model Variables', contains six checkboxes arranged in two columns. The second section, 'Export Model Information to XML File', contains a 'File:' label, a text input field, and a 'Browse...' button. At the bottom of the dialog box, there is a help icon (a circle with a question mark), a 'Cancel' button, and a 'Continue' button.

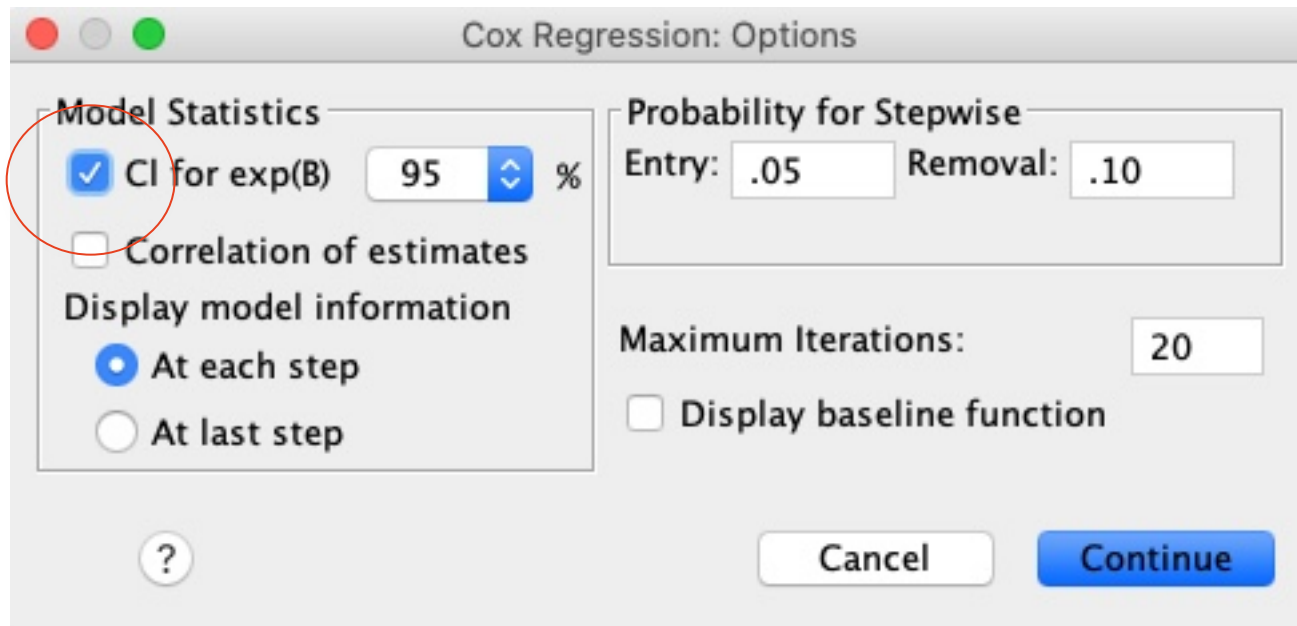
Cox Regression: Save		
Save Model Variables		
☒ Survival function	☒ Hazard function	☐ X * Beta
☐ Standard error of survival function	☒ Partial residuals	
☐ Log minus log survival function	☐ DfBeta(s)	
Export Model Information to XML File		
File:		Browse...
<input <="" td="" type="button" value="?"/> <td> Cancel </td> <td> Continue </td>	Cancel	Continue

2. Cox Regression Analysis

Step 1: Prepare the analysis

Additional step, not in the SPSS tutorial.

Options:



2. Cox Regression Analysis

Step 2: View and consider the output

Figure 1. Case processing summary

		N	Percent
Cases available in analysis	Event ^a	274	27.4%
	Censored	726	72.6%
	Total	1000	100.0%
Cases dropped	Cases with missing values	0	.0%
	Cases with negative time	0	.0%
	Censored cases before the earliest event in a stratum	0	.0%
	Total	0	.0%
Total		1000	100.0%

a. Dependent Variable: Months with service

The status variable identifies whether the event has occurred for a given case. If the event has not occurred, the case is said to be censored.

Censored cases are not used in the computation of the regression coefficients, but are used to compute the baseline hazard.

The case processing summary shows that 726 cases are censored. These are customers who have not churned.

2. Cox Regression Analysis

Step 2: View and consider the output

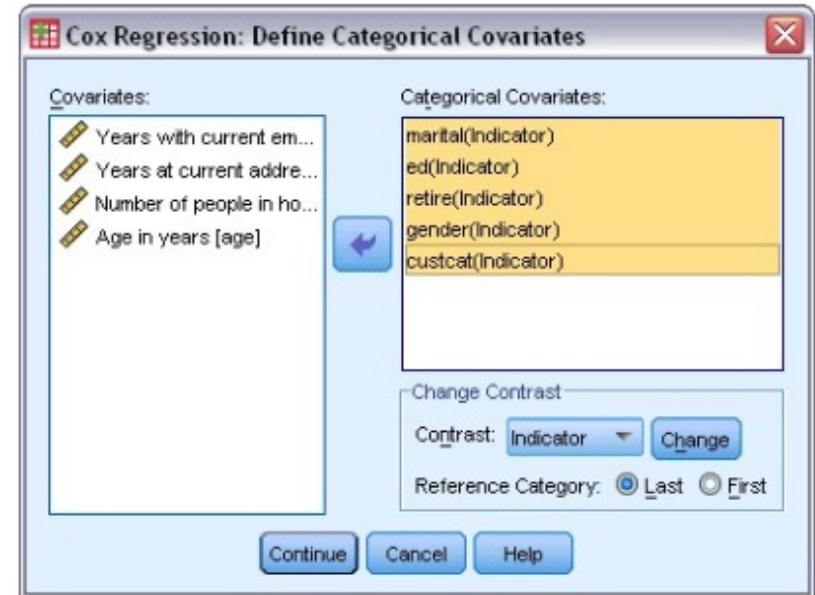
Figure 1. Categorical variable codings

		Frequency	(1) ^b	(2)	(3)	(4)
marital ^a	0=Unmarried	505	1			
	1=Married	495	0			
ed ^a	1=Did not complete high school	204	1	0	0	0
	2=High school degree	287	0	1	0	0
	3=Some college	209	0	0	1	0
	4=College degree	234	0	0	0	1
	5=Post-undergraduate degree	66	0	0	0	0
retire ^a	.00=No	953	1			
	1.00=Yes	47	0			
gender ^a	0=Male	483	1			
	1=Female	517	0			
custcat ^a	1=Basic service	266	1	0	0	
	2=E-service	217	0	1	0	
	3=Plus service	281	0	0	1	
	4=Total service	236	0	0	0	

a. Indicator Parameter Coding

b. The (0,1) variable has been recoded, so its coefficients will not be the same as for indicator (0,1) coding.

Figure 2. Categorical Covariate dialog box



2. Cox Regression Analysis

Step 2: View and consider the output

Figure 1. Omnibus tests for block 1

Step	-2 Log Likelihood	Overall (score)			Change From Previous Step			Change From Previous Block		
		Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square	df	Sig.
1 ^a	3383.793	132.522	1	.000	142.571	1	.000	142.571	1	.000
2 ^b	3331.588	161.504	2	.000	52.205	1	.000	194.776	2	.000
3 ^c	3295.644	178.903	3	.000	35.943	1	.000	230.720	3	.000
4 ^d	3295.688	174.203	2	.000	.044	1	.834	230.676	2	.000
5 ^e	3282.533	186.817	3	.000	13.155	1	.000	243.831	3	.000

a. Variable(s) Entered at Step Number 1: age

b. Variable(s) Entered at Step Number 2: employ

c. Variable(s) Entered at Step Number 3: address

d. Variable Removed at Step Number 4: age

e. Variable(s) Entered at Step Number 5: marital

Figure 2. Omnibus tests for Block 2

-2 Log Likelihood	Overall (score)			Change From Previous Step			Change From Previous Block		
	Chi-square	df	Sig.	Chi-square	df	Sig.	Chi-square	df	Sig.
3253.012	213.600	6	.000	29.521	3	.000	29.521	3	.000

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 3. Final variables in the equation

	B	SE	Wald	df	Sig.	Exp(B)
marital	.432	.123	12.358	1	.000	1.541
address	-.061	.009	49.768	1	.000	.940
employ	-.081	.010	67.141	1	.000	.922
custcat			28.506	3	.000	
custcat(1)	.121	.155	.612	1	.434	1.129
custcat(2)	-.574	.170	11.450	1	.001	.563
custcat(3)	-.658	.186	12.479	1	.000	.518

95% CI



The final model includes *marital*, *address*, *employ*, and *custcat*. To understand the effects of individual predictors, look at Exp(B), which can be interpreted as the predicted change in the hazard for a unit increase in the predictor.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 4. Variables not in the model

	Score	df	Sig.
age	.134	1	.715
ed	8.628	4	.071
ed(1)	2.869	1	.090
ed(2)	1.150	1	.283
ed(3)	.070	1	.791
ed(4)	5.519	1	.019
retire	.563	1	.453
gender	.128	1	.720
reside	1.195	1	.274

Variables left out of the model all have score statistics with significance values greater than 0.05.

However, the significance value for *ed*, while not less than 0.05, is very close.

It may be an interesting variable to pursue in further studies.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 1. Covariate means and patterns

	Mean	Pattern			
		1	2	3	4
age	41.684	41.684	41.684	41.684	41.684
marital	.505	.505	.505	.505	.505
address	11.551	11.551	11.551	11.551	11.551
ed(1)	.204	.204	.204	.204	.204
ed(2)	.287	.287	.287	.287	.287
ed(3)	.209	.209	.209	.209	.209
ed(4)	.234	.234	.234	.234	.234
employ	10.987	10.987	10.987	10.987	10.987
retire	.953	.953	.953	.953	.953
gender	.483	.483	.483	.483	.483
reside	2.331	2.331	2.331	2.331	2.331
custcat(1)	.266	1.000	.000	.000	.000
custcat(2)	.217	.000	1.000	.000	.000
custcat(3)	.281	.000	.000	1.000	.000

This table displays the average value of each predictor variable, plus a pattern for each level of *custcat*.

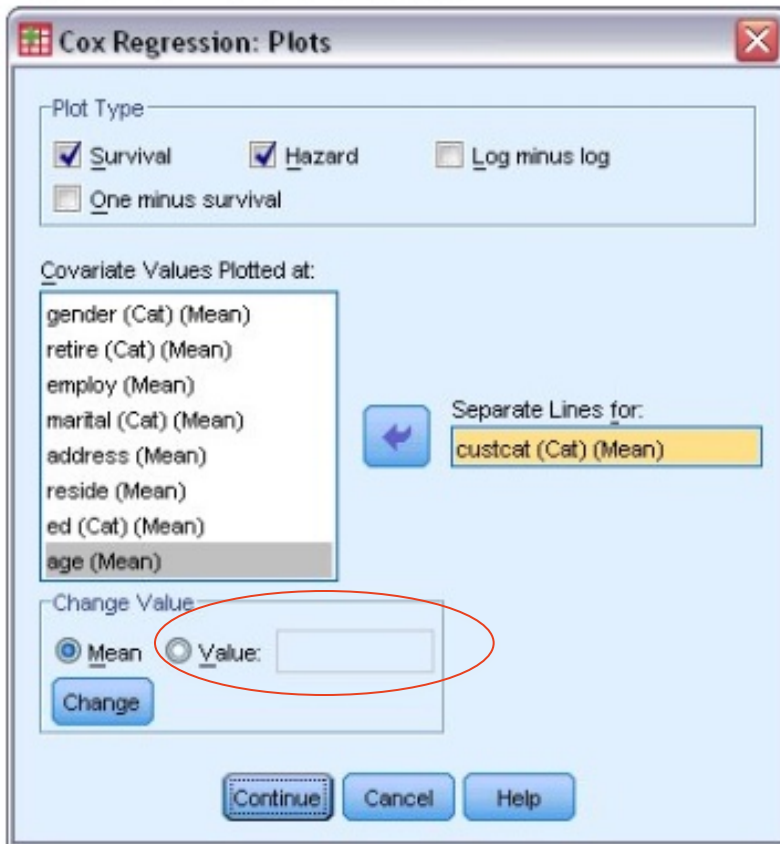
The four patterns correspond to each of the customer types, each with otherwise "average" predictor values.

This table is a useful reference when looking at the survival plots, which are constructed for the mean values and each covariate pattern.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 2. Plots dialog box



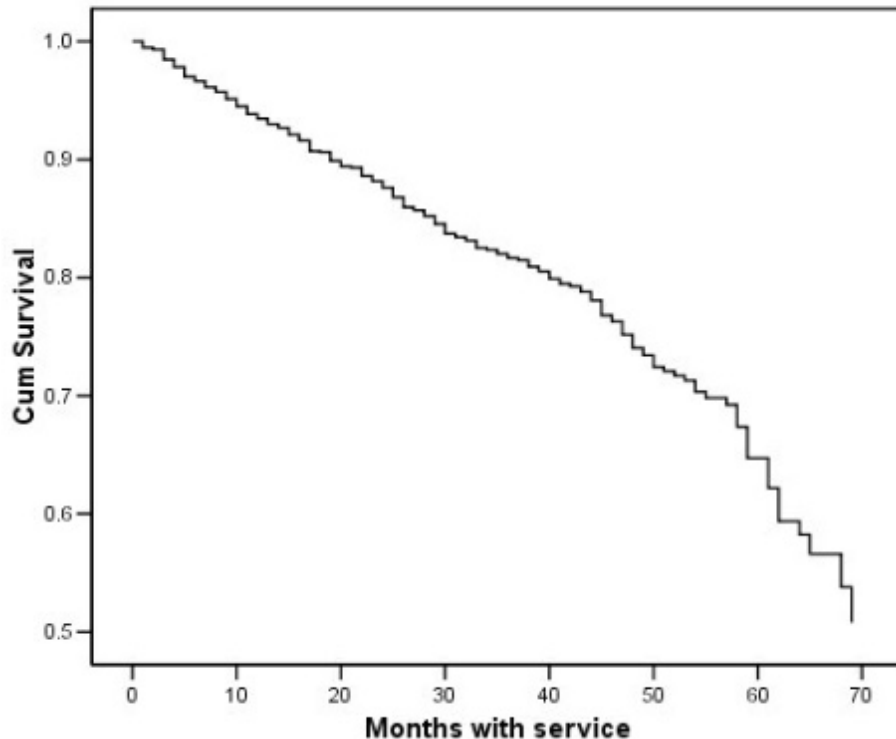
Note, however, that the "average" customer doesn't actually exist when you look at the means of indicator variables for categorical predictors. Even with all scale predictors, you are unlikely to find a customer whose covariate values are all close to the mean.

If you want to see the survival curve for a particular case, you can change the covariate values at which the survival curve is plotted in the Plots dialog box.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 1. Survival curve for "average" customer



The basic survival curve is a visual display of the model-predicted time to churn for the "average" customer. The horizontal axis shows the time to event.

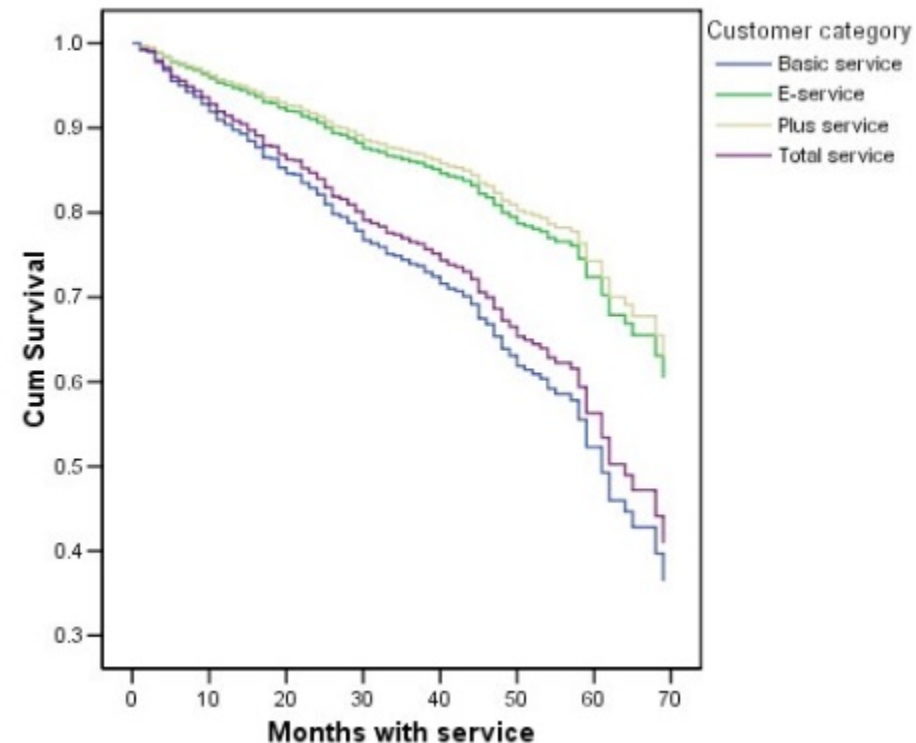
The vertical axis shows the probability of survival. Thus, any point on the survival curve shows the probability that the "average" customer will remain a customer past that time. Past 55 months, the survival curve becomes less smooth.

There are fewer customers who have been with the company for that long, so there is less information available, and thus the curve is blocky.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 2. Survival curves by customer category



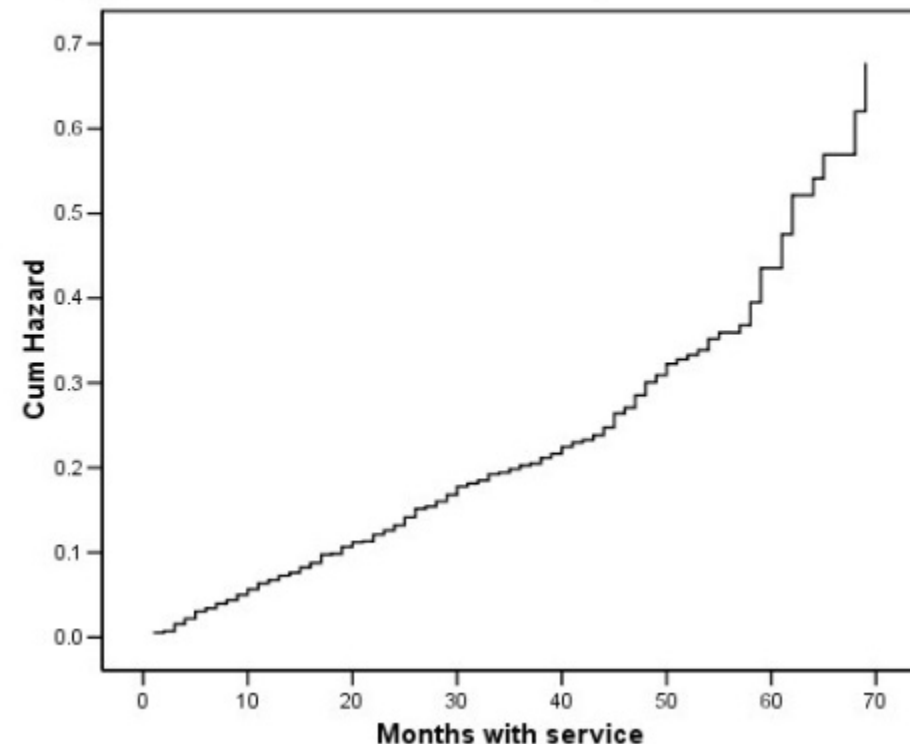
The plot of the survival curves for each covariate pattern gives a visual representation of the effect of *Customer category*.

Total service and *Basic service* customers have lower survival curves because, as you have learned from their regression coefficients, they are more likely to have shorter times to churn.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 1. Hazard curve for "average" customer



The basic hazard curve is a visual display of the cumulative model-predicted potential to churn for the "average" customer.

The horizontal axis shows the time to event.

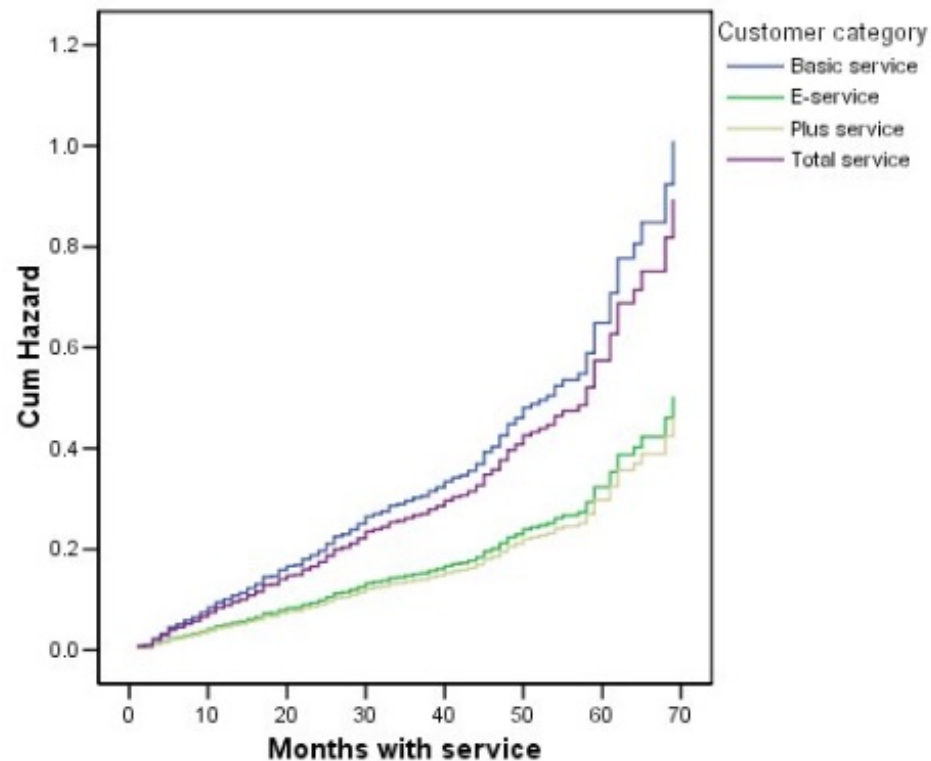
The vertical axis shows the cumulative hazard, equal to the negative log of the survival probability.

Past 55 months, the hazard curve, like the survival curve, becomes less smooth, for the same reason.

2. Cox Regression Analysis

Step 2: View and consider the output

Figure 2. Hazard curves by customer category



The plot of the hazard curves for each covariate pattern gives a visual representation of the effect of *Customer category*.

Total service and *Basic service* customers have higher hazard curves because, as you have learned from their regression coefficients, they have a greater potential to churn.

2. Cox Regression Analysis

Step 2: View and consider the output

Summary

Using Cox regression, you have found a suitable model for the time to churn. The use of separate blocks for fitting the model has allowed you to guarantee that *Customer category* would be in the final model, while still taking advantage of the stepwise techniques for choosing the other variables in the model.

To create this model, you included *Customer category* in the second block. Alternatively, the addition of *Customer category* to the model could have taken place in the first block, and the stepwise methods to choose the other variables in the second block. As it turns out, the final model is the same in this case, but that may not always be so. Thus, it is important to consider block order as well as block contents.

2. Cox Regression Analysis

Step 3: Checking the model fit (not in SPSS tutorial)

RESIDUALS

There are many types of residuals that could be examined:

1. Cox-Snell residuals (in SPSS called Hazard function, HAZ_1)
2. Modified Cox-Snell (two kinds!)
3. Martingale residuals
4. Deviance residuals
5. Schoenfeld residuals (in SPSS called partial residuals)
6. Score residuals

2. Cox Regression Analysis

Step 3: **RESIDUALS**

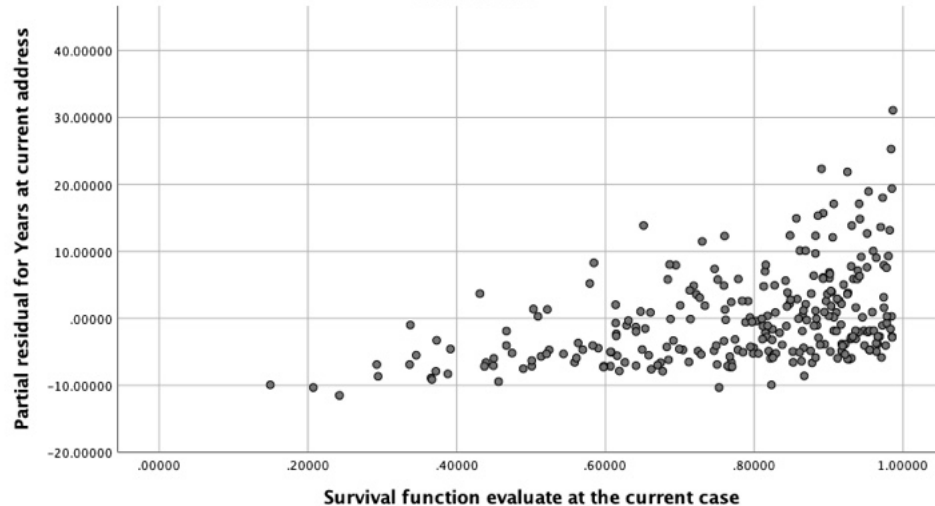
None of these residuals are as useful as those obtained in linear regression.

Residuals may display skewness and/or patterns in distribution even when the model fit is good.

2. Cox Regression Analysis

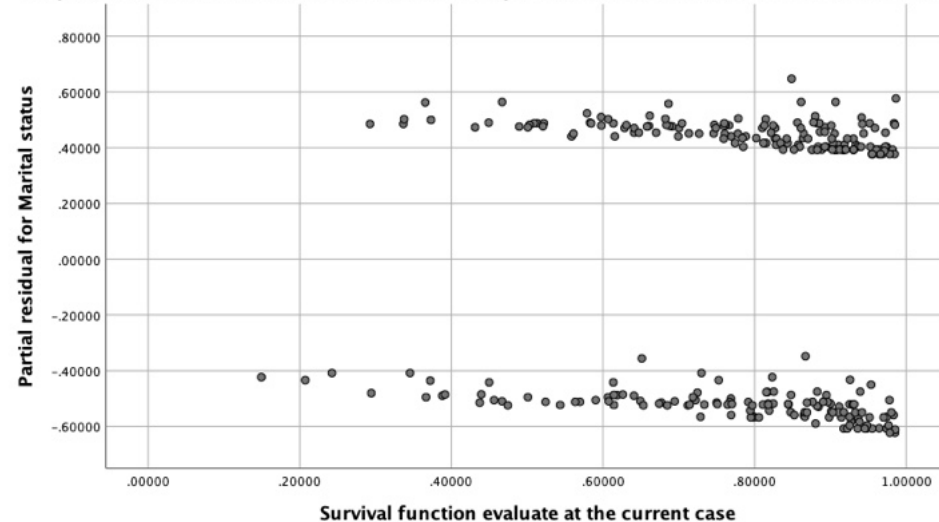
Step 3: Schoenfeld residuals (partial residuals) example

Simple Scatter of Partial residual for Years at current address by Survival function evaluate at the current case



Years at current address

Simple Scatter of Partial residual for Marital status by Survival function evaluate at the current case

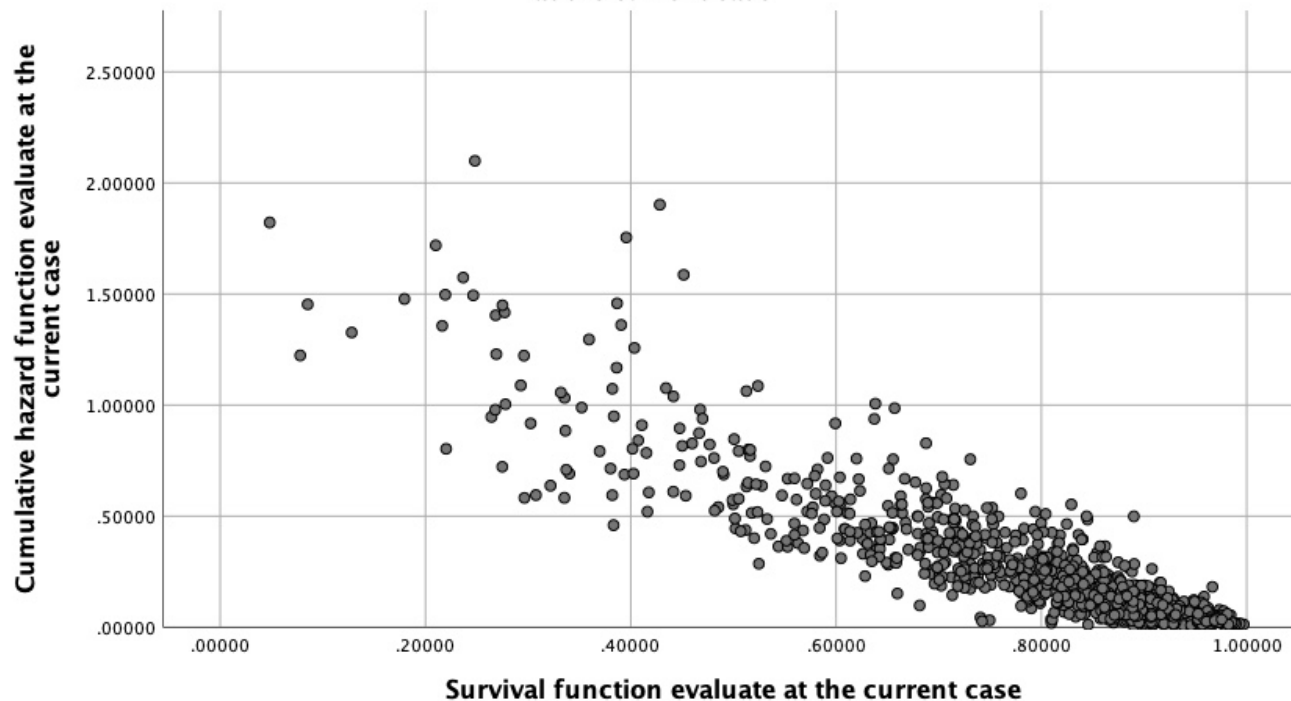


Marital status

2. Cox Regression Analysis

Step 3: Cox-Snell residuals (hazard function) example

Simple Scatter of Cumulative hazard function evaluate at the current case by Survival function evaluate at the current case



2. Cox Regression Analysis

Step 3: Computing other residuals

SPSS Syntax

* Compute the martingale and deviance residuals.

* Note that status variable name is “churn”.

```
compute martingale=(churn=1)-HAZ_1.
```

```
compute deviance = SQRT(-2*(martingale+(churn=1)*ln((churn=1)-  
martingale))).
```

```
if martingale<0 deviance=-deviance.
```

```
formats martingale deviance (F8.5).
```

```
execute.
```

* Create case ID variable.

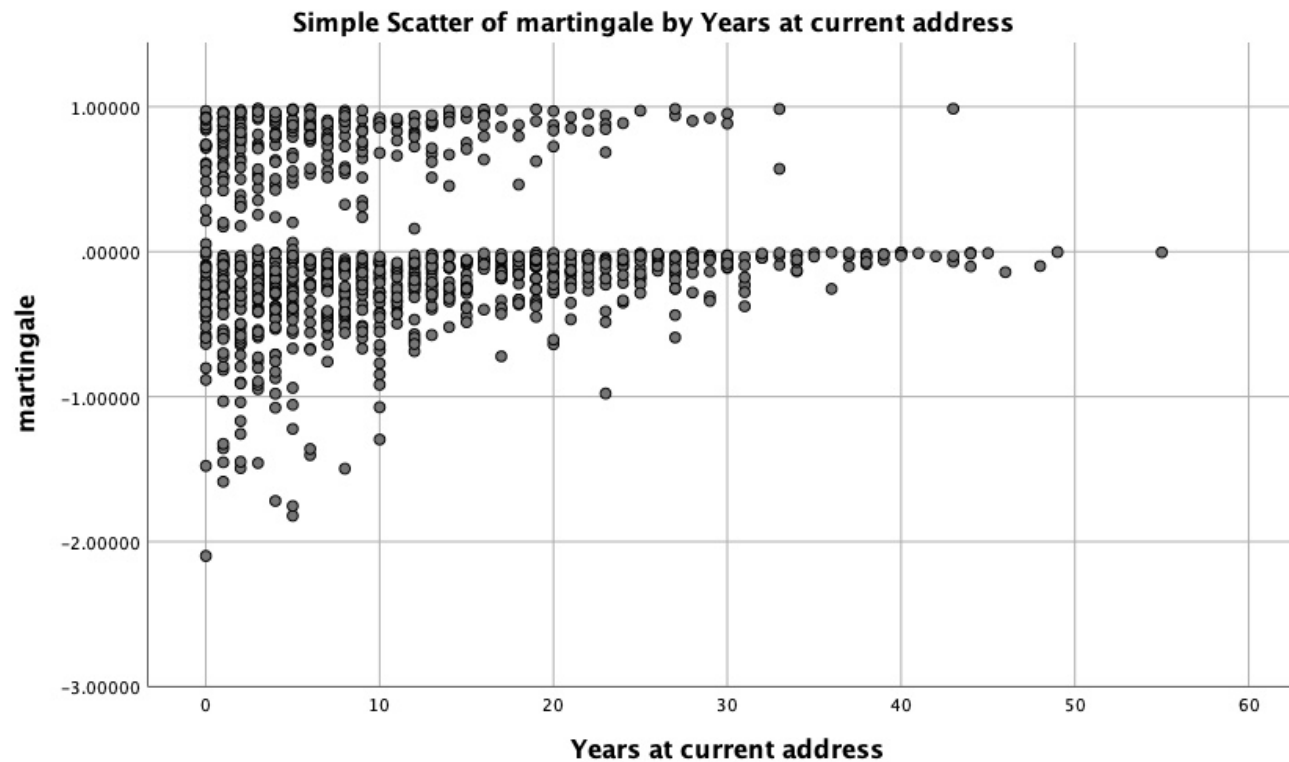
```
COMPUTE id=$CASENUM.
```

```
FORMAT id (F8.0).
```

```
EXECUTE.
```

2. Cox Regression Analysis

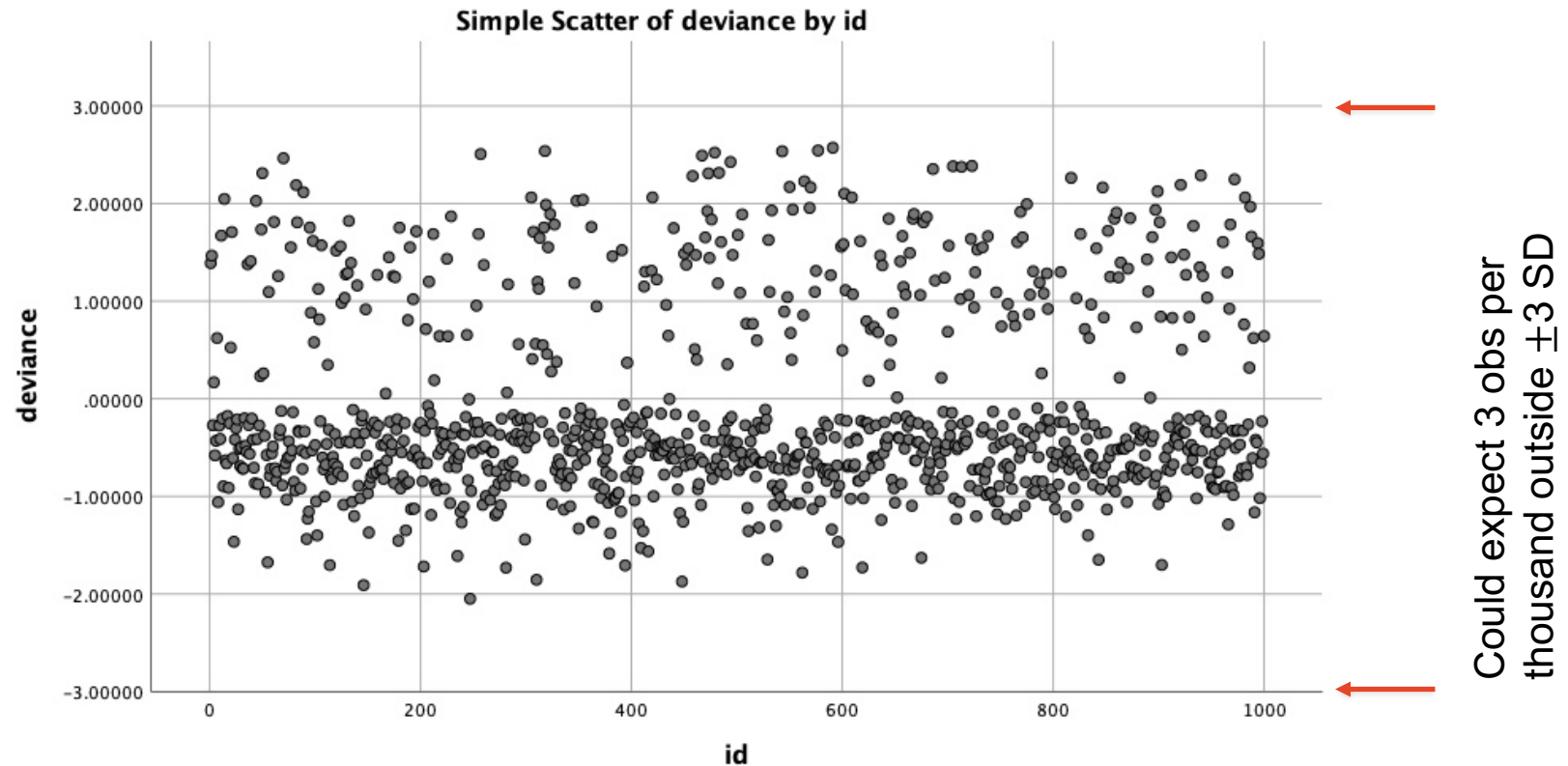
Step 3: Martingale residuals example



- Censored cases have negative residuals
- Still exhibits skewed distribution

2. Cox Regression Analysis

Step 3: Deviance residuals example



- Should be symmetric around zero
- Can identify outliers
- Standard Deviation = 1.0

2. Cox Regression Analysis

Step 4: Reporting

- Descriptive Statistics
- Model building method – eg Stepwise Forward LR in SPSS
- Validity of assumptions
- Judgements made in the model building
- Hazard Ratios with confidence intervals
- Survival curves, hazard curves

2. Cox Regression Analysis – Glaucoma example

A researcher from the Save Sight Institute was interested in the incidence of secondary glaucoma following paediatric cataract surgery.

1. Data Preparation
2. Exploratory Data Analysis
3. Consider different functional forms of covariates (eg. Age continuous or binary, Surgical technique 2, 3 or 4 categories)
4. Univariate Cox Regression models

2. Cox Regression Analysis – Glaucoma example

A researcher from the Save Sight Institute was interested in the incidence of secondary glaucoma following paediatric cataract surgery.

5. Use $-2\log L$ and AIC to build a multivariate model (following the process given by Collett). Consider interaction terms as well.
6. Compare with Forward Stepwise LR process.
7. Report best (2 in this case) multivariate model(s).
8. Report SPSS syntax for all steps to allow reproducibility of analysis

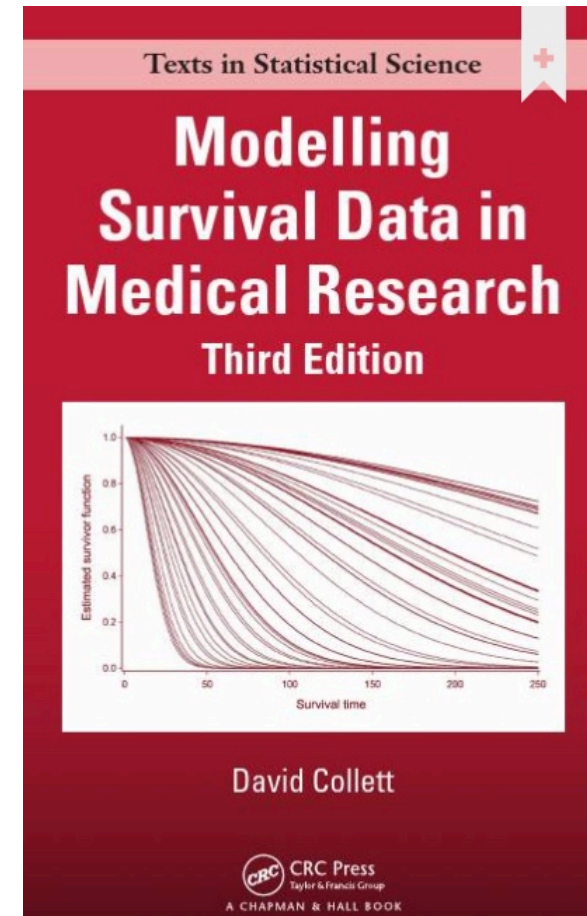
Survival Analysis with SPSS

Other resources:

Video: LinkedIn Learning – Specialized statistics with Stata/Survival Analysis

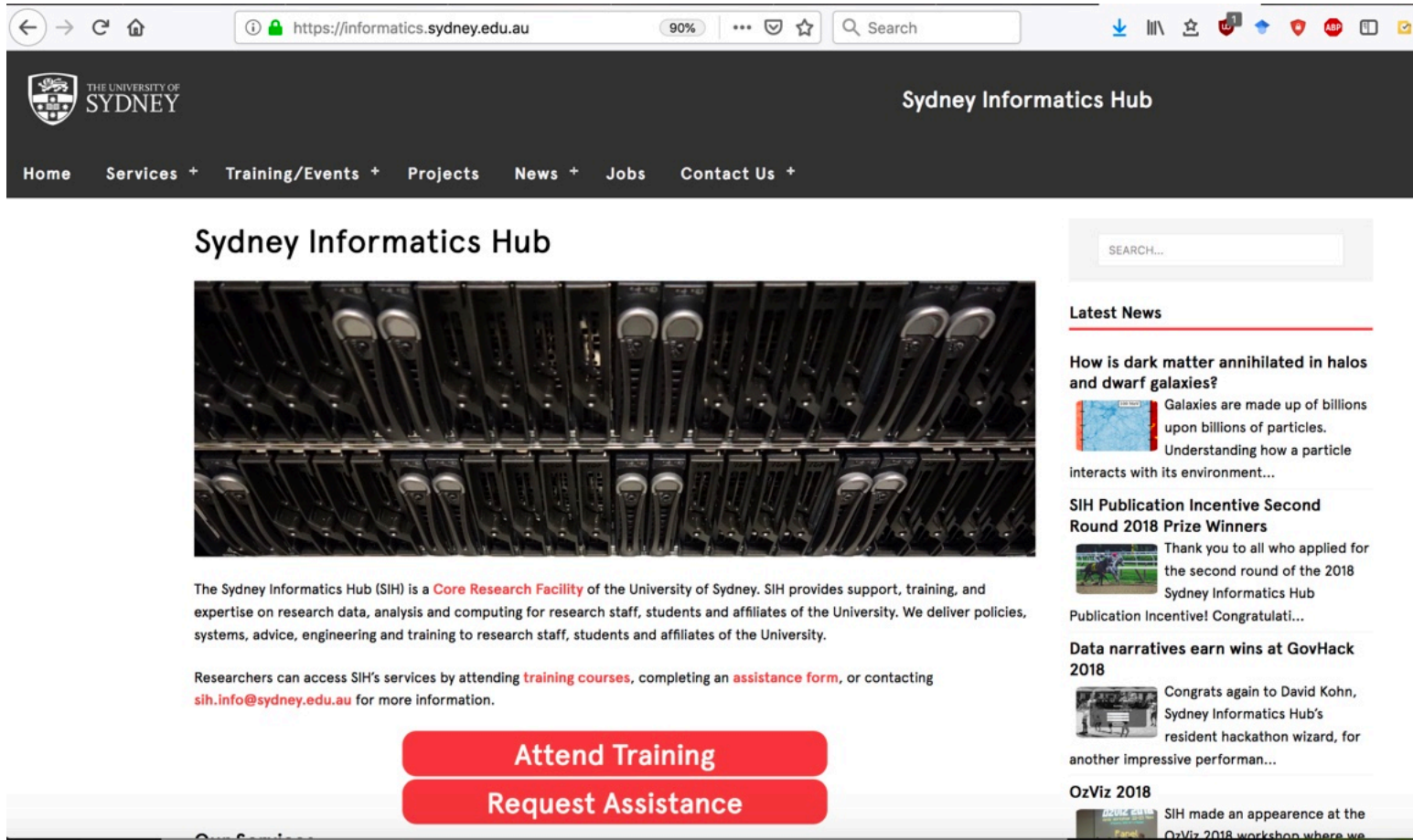
Book: Collett, D. *Modelling Survival Data in Medical Research*

Sydney Informatics Hub – support for HDR students and researchers affiliated with the University of Sydney



Sydney Informatics Hub - website

Current website (until early Dec)



The screenshot shows the Sydney Informatics Hub website. The browser address bar displays <https://informatics.sydney.edu.au>. The website header includes the University of Sydney logo and the text "Sydney Informatics Hub". A navigation menu contains links for Home, Services, Training/Events, Projects, News, Jobs, and Contact Us. The main content area features a large image of server racks and the heading "Sydney Informatics Hub". Below the image, a paragraph describes the hub as a Core Research Facility. A search bar is located in the top right. A "Latest News" section on the right lists three articles: "How is dark matter annihilated in halos and dwarf galaxies?", "SIH Publication Incentive Second Round 2018 Prize Winners", and "Data narratives earn wins at GovHack 2018". At the bottom, two red buttons are visible: "Attend Training" and "Request Assistance".

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OzViz 2018

SIH made an appearance at the OzViz 2018 workshop where we

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Survival Analysis with SPSS

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Do you have any other feedback on the course or how it was run (things you liked, didn't like, things that were missing, any other comments)?

Survival Analysis using SPSS

- Thank you for your interest and attention
- Questions and comments welcome

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